Strategic Investors and Exchange Rate Dynamics

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Abstract

We study how the exchange rate dynamics are influenced by the presence of heterogeneous investors with varying degrees of price impact. Foreign exchange markets are historically characterized by a high degree of concentration, with a relatively small number of investors holding substantial currency positions that give them the potential to influence currency prices. We develop a monetary model of exchange rate determination that incorporates heterogeneous investors with different degrees of price impact. Our model predicts that the presence of price impact amplifies the exchange rate's response to non-fundamental shocks while dampening its response to fundamental shocks. As a result, investors' price impact contributes to the disconnect of exchange rates from fundamentals, increases the excess volatility of exchange rates, and makes excess return more predictable. We provide empirical evidence in line with our theoretical predictions, using data on trading volume concentration from the U.S. Commodity Futures Trading Commission (CFTC) foreign exchange rate market for 10 currencies spanning from 2006 to 2016.

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1 Introduction

Three well-known puzzles in international economics are the limited explanatory power of macroeconomic fundamentals in accounting for exchange rate fluctuations (the exchange rate determination puzzle), the excessive volatility of exchange rates relative to fundamentals (the excess volatility puzzle), and the failure of uncovered interest parity (the UIP puzzle) (Meese and Rogoff, 1983; Obstfeld and Rogoff, 2000; Fama, 1984).[1] Recent evidence from the microstructure approach to exchange rates suggests that investor heterogeneity plays a crucial role in understanding exchange rate dynamics and determination. For example, these puzzles can be explained by the rational confusion arising from information heterogeneity (Bacchetta and Van Wincoop, 2006). Similarly, exchange rate behavior is linked to order flow, which, in turn, is associated with the heterogeneity among investors (Lyons et al., 2001; Evans and Lyons, 2006). This paper investigates how the presence of large investors who internalize the impact of their trading on currency prices can potentially influence exchange rate fluctuations.

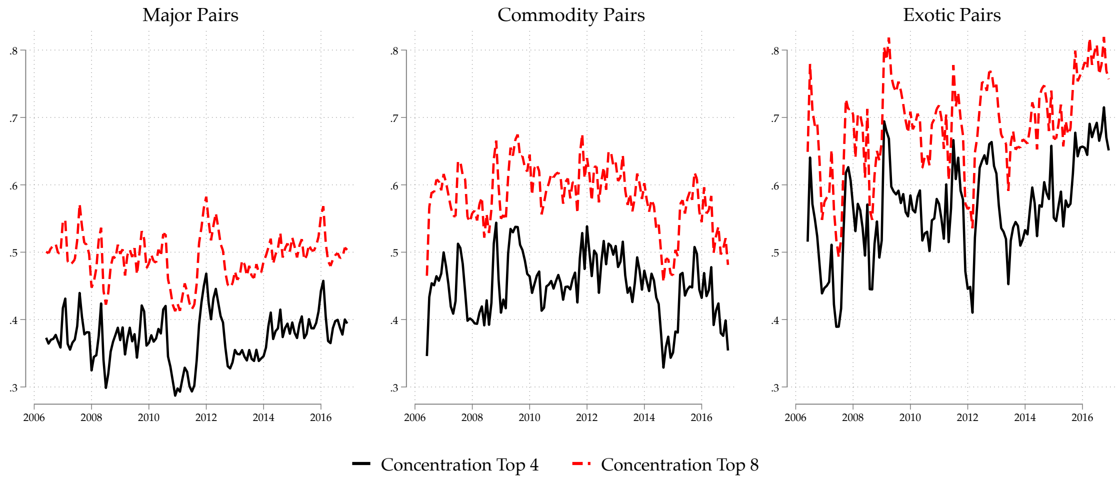
Foreign exchange markets are historically characterized by an high degree of concentration, with a relatively small number of investors holding a substantial presence in the markets through their currency positions. Figure 1 reports the average concentration ratios by currency groups (major and non-major currencies), computed as the share of net open interest positions held by the largest four and eight investors operating the principal foreign exchange markets from 2006 to 2016. Two facts stand out. Firstly, the eight (four) largest entities collectively held approximately 50% to 70% (40% to 60%) of the open interest positions in the foreign exchange market. Secondly, non-major USD currency pairs are notably more concentrated, by about 20%, compared to major ones.[2] The high concentration measured here qualitatively aligns with other pieces of evidence, such as the BIS Triennial Survey of Foreign Exchange Markets or the NY FED OTC Foreign Exchange Market Survey (Appendix Figure A.5 and Table A.5). Figure 1 suggests the potential for individual investors to influence currency prices (Kyle, 1985).[3]

[1] Meese and Rogoff (1983) show that macroeconomic models have lower predictive power compared to a random walk model. Similarly, Obstfeld and Rogoff (2000) show that exchange rates exhibit significantly more fluctuations than their underlying fundamentals.

[2] Appendix A provide more information on the data. Figure A.3 in Appendix A shows the concentration ratios for all individual currencies from 2006 to 2016. The Brazilian Real, Russian Ruble, and New Zealand Dollar are the currency pairs with the highest concentration, while the Euro and Canadian Dollar exhibit the lowest. Figure A.4 in Appendix A shows qualitatively similar patterns when concentration is measured by the number of entities trading each currency (on average, 10 to 25 entities actively trade, with more traders active in major currency markets).

[3] This potential is not just theoretical. The 2013 London FX fixing scandal, the 2015 Swiss franc peg removal, and the 2016 British pound flash crash all demonstrate how concentrated trading flows of large traders can significantly impact currency markets, leading to substantial price movements and market disruptions. Despite significant institutional reforms implemented in 2015, there are indications that market manipulation may not have completely ceased (Osler, 2014; Osler et al., 2016; Cochrane, 2015).

Figure 1: Market Concentration – U.S. CFTC



Notes: The figure shows the average concentration ratio of net open interest positions help by asset managers, institutional investors, and leveraged funds across currencies, divided between major and non-major currency groups. We consider the share held by the eight and the four largest entities in each market. Concentration ratios are computed on 'Net Position', meaning that are calculated after offsetting each trader's long and short positions. Major currency pairs consist of the United States Dollar paired with the Euro, British Pound, Japanese Yen, and Swiss Franc. Non-Major currency pairs include the United States Dollar paired with the Australian Dollar, Canadian Dollar, New Zealand Dollar, Mexican Peso, Brazilian Real, and Russian Ruble. The data is sourced from the U.S. Commodity Futures Trading Commission (CFTC) and spans from 2006 to 2016, with quarterly averages for each currency pair. Appendix A provides additional details regarding the data used.

We embed heterogeneity in price impact into a two-country, dynamic monetary model of exchange rate determination to analyze the potential effects of strategic behavior on exchange rate fluctuations. In our theoretical framework, investors face an international portfolio choice problem with noise shocks. Departing from the conventional assumption of price-taking investors, we introduce a continuum of investors with varying degrees of price impact. One group of investors is atomistic and competitive, acting as price takers, while the remaining group consists of a finite number of strategic investors with non-zero market power, who behave oligopolistically and internalize the effects of their trading decisions on equilibrium prices. In equilibrium, the exchange rate is a weighted average of current and expected future fundamentals, such as interest rate differentials, and noise shocks, with the weights depending on market structure and the average size of strategic investors.

We present a set of theoretical predictions on how strategic investors influence exchange rate dynamics. First, the presence of strategic investors amplifies the exchange rate's response to noise shocks while dampening its response to fundamental shocks. In the model, a positive noise shock decreases the residual demand for foreign assets, driving up their price and the exchange rate, without changes in fundamentals. Conversely, a negative fundamental shock increases the excess returns on foreign assets, resulting in an appreciation of the foreign currency. Strategic investors, aware of their price impact, adjust their trading positions less aggressively, which amplifies the effect of noise shocks and weakens the impact of fundamental shocks on the exchange rate.

Our theoretical framework predicts that heterogeneity in price impact can contribute to exchange rate predictability, the exchange rate disconnect and the excess volatility of the exchange rate relative to fundamentals. First, in the model, systematic deviations from Uncovered Interest Parity (UIP) arise due to a non-zero net supply of foreign assets. Strategic behavior amplifies UIP deviations, making the Fama coefficient more negative and reflecting a stronger influence of interest rate differentials on excess returns and risk premiums. Second, the presence of strategic investors leads to a reduction in the information loading factor of the exchange rate (reduced informativeness), meaning that the exchange rate provides less information about underlying fundamentals. Lastly, by increasing the importance of the noise component in exchange rate dynamics, the presence of strategic investors contributes to the heightened volatility of exchange rates compared to underlying fundamental factors, as fundamental factors exhibit lower volatility compared to noise shocks.

We provide empirical evidence consistent with the theoretical predictions of the model using a panel of 10 currencies spanning from 2006 to 2016. We leverage the variation in market concentration across currencies and time to test the implications of our theory. For this purpose, we combine daily exchange rate data with currency-level concentration data from the U.S. Commodity Futures Trading Commission (CFTC). These data cover both major and non-major currencies and provide standardized measures of trading activity over a relatively long period, but capture only a subset of market participants in the U.S. futures market and therefore provide an informative but necessarily incomplete measure of concentration in global foreign exchange markets. Our empirical results show that, consistent with the theoretical predictions, a currency traded in a market with a 10% higher share of strategic investors exhibits an 18% lower predictive power compared to the average predictive power in the data. Similarly, a currency traded in a market with a 10% higher share of strategic investors exhibits an excess volatility ratio that is 12% higher than the average ratio.

We assess the potential impact of strategic investors on exchange rates by calibrating a benchmark model with strategic investors and comparing it to a competitive model. Under reasonable parameters, we show that the presence of strategic investors may significantly influence exchange rate disconnect. However, while strategic investors and heterogeneity in price impact may theoretically exacerbate the forward premium and excess volatility puzzles, they explain only a moderate portion of the puzzles. Abstracting from the heterogeneity in price impact increases the connection between fundamentals and exchange rates by 12%, while it accounts for a moderate increase in excess return predictability and exchange rate volatility of approximately 6%. In the sensitivity analysis, we show that the impact of investors' heterogeneity in price changes on exchange rate dynamics can substantially increase for stronger magnitudes of strategic behavior, supporting the idea that investor-level features may be quantitatively relevant for aggregate dynamics.

1.1 Related literature

Our work contributes to the microstructure approach to exchange rates by focusing on the heterogeneity of investors' price impact. Recent evidence from this literature highlight the importance of investor heterogeneity in understanding exchange rate dynamics and determination. For instance, the exchange rate determination puzzle, the excess predictability puzzle and the excess volatility puzzle can be explained by the rational confusion resulting from information heterogeneity among investors (Bacchetta and Van Wincoop, 2006; Candian and De Leo, 2022; Stavrakeva and Tang, 2020). Furthermore, exchange rate behavior is linked to order flow, which, in turn, is associated with the heterogeneity among investors (Lyons et al., 2001; Evans and Lyons, 2006). However, despite extensive evidence that foreign exchange rate markets are highly concentrated and atomistic price-taking investors are hardly realistic, the literature has ignored the potential heterogeneity in price impact (Osler, 2014; Osler et al., 2016; Cochrane, 2015). A notable exception is the work in Corsetti et al. (2004) and Corsetti et al. (2002), which theoretically studies the role that large investors have in speculative attacks in the foreign exchange markets. Differently to them, we focus on exchange rate determination and puzzles by incorporating heterogeneity in price impact, drawing on the modeling approach of Kyle (1989) and Kacperczyk et al. (2025), which has not been previously applied in the context of exchange rate markets.

This paper contributes to the rich literature on the determination and dynamics of exchange rates in the presence of frictions. Prior work explores various types of frictions, including informational frictions (Evans and Lyons, 2002; Bacchetta and Van Wincoop, 2006), infrequent portfolio adjustment (Bacchetta and Van Wincoop, 2010, 2019), imperfect and frictional markets (Gabaix and Maggiori, 2015; He and Krishnamurthy, 2013). To the best of our knowledge, our work is the first to specifically focus on this aspect of the market structure – the presence of strategic investors and heterogeneity in price impact – for the determination of the exchange rate.

This paper also relates to the vast literature attempting to explain major puzzles in international economics, both theoretically and empirically. We contribute by providing a new rationale, based on strategic behavior and price impact, for the failure of macroeconomic fundamentals to predict exchange rates and the large volatility of the exchange rate relative to fundamentals (Meese and Rogoff, 1983; Obstfeld and Rogoff, 2000; Engel and Zhu, 2019). We also show that the presence of strategic behavior and excess predictability interact (Fama, 1984). Although we do not propose novel explanations for UIP deviations, the presence of strategic investors can account for currency level differences in UIP deviations. Moreover, we empirically study cross-currency differences in exchange rate puzzles and dynamics, which have been relatively unexplored, and find that different levels of price impact can explain cross-currency differences in a panel of 10 currencies.

The rest of the paper is organized as follows. Section 2 introduces the theoretical framework and explains the fundamental mechanism of strategic behavior. In Section 3, we discuss the main implications for the dynamics of the exchange rate and provide empirical evidence that supports the theoretical predictions. Finally, Section 4 presents the conclusion. Any proofs, derivations, and robustness analyses that were omitted can be found in the Appendices.

2 A Monetary Model with Strategic Investors

We propose a framework that incorporates strategic behavior in the spirit of Kyle (1989) and Kacperczyk et al. (2025) into a standard two-country, discrete time, general equilibrium monetary model of exchange rate determination (Mussa, 1982; Jeanne and Rose, 2002). We study the implications of strategic behavior in the context of a monetary model, as it is well-suited for analyzing short-term exchange rate variations. To isolate the key mechanisms, we assume that agents have rational expectations about the dynamics of the exchange rate.[4].

2.1 Basic Set-up

There are two economies, Home and Foreign, both producing the same good. Variables referring to Foreign are indicated with a star. We assume that purchasing power parity holds, so that:

$$p_t = p_t^\star + s_t, \tag{1}$$

where s_t is the log nominal exchange rate, p_t ($p_t^\star$) the log price level in the Home (Foreign) country. The exchange rate is defined as the value of the foreign currency in term of domestic currency, and an increase in the exchange rate reflects an appreciation of the foreign currency. There are three assets: one-period nominal bonds issued by both Home and Foreign with interest rates i_t and $i_t^\star$, respectively, and a risk-free technology with fixed real return r. The latter is infinitely supplied while bonds are in fixed supply in their respective currency. We follow Bacchetta and Van Wincoop

[4]Qualitatively similar predictions can be derived in a model extension that allows for dispersed information, following Bacchetta and Van Wincoop (2006) and Nimark (2017).

(2010) and assume asymmetric monetary rules between the two countries. The Home central bank commits to a constant price level, $p_t = 0$, which implies that the domestic interest rate is equal to the risk free technology, $i_t = r$. On the other hand, the monetary policy in Foreign is stochastic, $i^\star_t = -u_t$ where

$$u_t = \rho_u u_{t-1} + \sigma_u \varepsilon^u_t, \quad \varepsilon^u_t \sim N(0,1), \quad \rho_u \in (0,1) \tag{2}$$

is the Foreign monetary policy shock. Thus, the interest rate differential is defined as

$$i_t - i^\star_t = u_t + r, \tag{3}$$

implying that the dynamics of the exchange rate are solely influenced by the monetary policy of the Foreign country.[5] In our model, we refer to a shock in the Foreign monetary policy as a fundamental shock.

Investors. There is a continuum of investors of mass one in the Home country, and an infinitesimal mass of investors in the Foreign country. Since the mass of Foreign investors is infinitesimal, their portfolio positions do not affect the equilibrium exchange rate, which is determined entirely by Home investors, whom we model throughout.[6] We assume overlapping generations of investors, each living for two periods and making a single portfolio decision, deriving utility only from end-of-life wealth; this abstracts from intertemporal saving (Bacchetta and Van Wincoop, 2006, 2010). Each investor is born with an exogenous endowment ω, which it allocates between nominal bonds and the risk-free technology.

Home investors are of two types, competitive (C) and strategic (S). A mass $1-\lambda$ are atomistic price takers, while the remaining mass λ consists of a finite number N of strategic investors, each of positive mass λ_i with $\sum_{i=1}^N \lambda_i = \lambda$. Strategic investors operate as an oligopoly: they choose their bond positions simultaneously, each taking the other strategic investors' positions as given while internalizing the effect of its own trades on the equilibrium exchange rate, which is determined by market clearing.

Investor j chooses its foreign-bond holdings b^j_t to maximize mean–variance utility over next-

[5] Bacchetta and Van Wincoop (2010) specify a simplified Wicksellian rule of the form $i^\star_t = \psi(p^\star_t - \bar{p}^\star) - u_t$ where ψ is set equal to zero, consistent with the low estimates of ψ reported by Engel and West (2005). Bacchetta and Van Wincoop (2010) show that an exogenous interest rate rule, as in our case, does not compromise the existence of a unique stochastic steady state for the exchange rate.

[6] This assumption does not imply that Foreign assets are economically irrelevant: the Foreign interest rate is determined by an exogenous stochastic monetary policy process and affects the relative returns faced by Home investors, thereby influencing their portfolio decisions.

period wealth, w^j_{t+1}:

$$\max_{b^j_t} \quad \mathbb{E}^j_t\left(w^j_{t+1} \mid \Omega^j_t\right) - \frac{\rho}{2}\,\mathrm{Var}^j_t\left(w^j_{t+1} \mid \Omega^j_t\right) \tag{4}$$

$$\text{s.t.} \quad w^j_{t+1} = (\omega - b^j_t)\, i_t + (i^\star_t + s_{t+1} - s_t)\, b^j_t \tag{5}$$

where b^j_t denotes foreign-bond holdings, $\omega - b^j_t$ the remaining endowment allocated to domestic bonds, ρ the coefficient of risk aversion, and Ω^j_t investor j's information set at time t. The domestic and foreign returns, i_t and $i^\star_t + s_{t+1} - s_t$, are expressed in log-linearized form; under PPP and the monetary-policy assumptions above, $p^\star_t = -s_t$, so both are real returns, and the two assets differ only in that the foreign return is stochastic.[7] Agents have symmetric rational expectations about the dynamics of the exchange rate, so $\Omega^j_t = \Omega_t$.

Noise Traders. Besides the competitive and strategic investors, the model includes noise traders, whose demand for foreign bonds is exogenous. As is standard, their presence allows the model to match key empirical moments of exchange rates, such as volatility, disconnect, and deviations from UIP (Kyle, 1989; Bacchetta and Van Wincoop, 2006, 2010). Following Bacchetta and Van Wincoop (2010), noise-trader demand is given by

$$X_t = (\bar{x} + x_t)\bar{W}, \tag{6}$$

where $\bar{W}$ is the steady-state aggregate financial wealth in the Home economy, $\bar{x}$ is a constant, and x_t follows the exogenous AR(1) process

$$x_t = \rho_x x_{t-1} + \sigma_x \varepsilon^x_t, \quad \varepsilon^x_t \sim N(0,1), \quad \rho_x \in (0,1). \tag{7}$$

In the stochastic steady state, the demand for foreign assets absorbed by noise traders equals $\bar{x}\bar{W}$. Deviations from it are driven by x_t, a noise shock orthogonal to the fundamental shock u_t in Equation (2). A positive shock to x_t raises the desire for foreign assets, leading the foreign currency to appreciate without any movement in the interest-rate differential.

[7] $p_t = 0$ implies $i_t = r$. Similarly, $p^\star_t = -s_t$ implies that the return on foreign bonds, $i^\star_t + s_{t+1} - s_t$, is expressed in real terms as well.

Market Clearing. Given the optimal demands of competitive investors, strategic investors, and noise traders, the foreign-bond market clears when total demand equals supply:[8]

$$(1-\lambda)\, b_t^C + \sum_{i=1}^{N} \lambda_i\, b_t^{S,i} + X_t = B\,(1+s_t)\,, \tag{8}$$

where the left-hand side is the total demand for foreign bonds — from competitive investors, $(1-\lambda)\, b_t^C$; strategic investors, $\sum_{i=1}^{N} \lambda_i b_t^{S,i}$; and noise traders, X_t — and the right-hand side is the supply of foreign bonds, B, valued in domestic currency: because foreign bonds are denominated in foreign currency, their domestic-currency value rises with the exchange rate.[9]

2.2 Equilibrium and Mechanism

Definition 1 (Equilibrium)**.** Given the exogenous processes $\{u_t\}$ and $\{x_t\}$ for the fundamental and noise shocks, a rational-expectations equilibrium is a sequence of Home investors' portfolio holdings, b_t^C and $\{b_t^{S,i}\}_{i=1}^N$ and exchange rates s_t such that:

(i) Competitive investors choose b_t^C to solve (4)–(5) taking s_t as given;

(ii) Each strategic investor i chooses $b_t^{S,i}$ to solve (4)–(5), taking the other strategic investors' positions $\{b_t^{S,j}\}_{j\neq i}$ as given and internalizing its price impact $\partial s_t/\partial b_t^{S,i}$;

(iii) The foreign-bond market clearing condition in Equation (8) holds;

(iv) Expectations are rational implying that the conditional moments on s_{t+1} coincide with those implied by the equilibrium law of motion of the exchange rate.

To characterize the equilibrium, we first derive Home investors' optimal allocations and then impose the market-clearing condition in Equation (8) to solve for price impact and the exchange rate dynamics. A detailed derivation of the market equilibrium can be found in Appendix B.1.

Foreign Assets Demand. Investors' demand schedule and portfolio allocation vary depending on their type, competitive (C) and strategic (S). Strategic investors internalize the effects that

[8] The market clearing for the domestic bond is not explicitly considered because domestic bonds are perfectly substitutable with the risk-free technology, which is infinitely supplied. Furthermore, in a monetary model, a market clearing condition for the money market would also be required. Bacchetta and Van Wincoop (2006) and Bacchetta and Van Wincoop (2010) assume that investors generate a money demand (independent of their portfolio decision) and that money supply accommodates it under the exogenous rule for interest rates. We do not explicitly model the money market in order to limit notation, leaving it in the background.

[9] To obtain a closed-form solution for the equilibrium exchange rate, we take a linear approximation of the foreign-asset supply Be^{s_t} around $s_t = 0$.

their demand has on equilibrium prices—more precisely, on the equilibrium exchange rate—, while competitive investors do not.

Substituting the budget constraint in Equation (5) into the objective function in Equation (4), and noting that s_{t+1} is the only source of date-$(t+1)$ uncertainty, investor j chooses b_t^j to maximize

$$\omega\, i_t + b_t^j \left(\mathbb{E}_t s_{t+1} - s_t + i_t^\star - i_t\right) - \frac{\rho}{2}\,(b_t^j)^2\,\sigma_t^2, \qquad \sigma_t^2 \equiv \mathrm{Var}_t(s_{t+1}).$$

The first-order condition sets the expected excess return equal to the marginal cost of holding foreign bonds,

$$\mathbb{E}_t s_{t+1} - s_t + i_t^\star - i_t = \left(\rho\,\sigma_t^2 + \frac{\partial s_t}{\partial b_t^j}\right) b_t^j, \tag{9}$$

which comprises the marginal cost of risk, $\rho\,\sigma_t^2\, b_t^j$, and, for strategic investors, the cost of moving the exchange rate against their own position, $b_t^j\,\partial s_t/\partial b_t^j$. Strategic investors internalize this contemporaneous price effect, taking the conditional mean and variance of s_{t+1} as given, whereas competitive investors take s_t as given, so $\partial s_t/\partial b_t^C = 0$. Solving Equation (9) for b_t^j yields the demand schedules

$$b_t^j = \begin{cases} \dfrac{\mathbb{E}_t s_{t+1} - s_t + i_t^\star - i_t}{\rho\sigma_t^2}, & j = C, \\[2ex] \dfrac{\mathbb{E}_t s_{t+1} - s_t + i_t^\star - i_t}{\rho\sigma_t^2 + \frac{\partial s_t}{\partial b_t^S}}, & j = S. \end{cases} \tag{10}$$

We focus on a stochastic steady state in which σ_t^2 is time-invariant. Demand rises with the expected excess return $\mathbb{E}_t q_{t+1} \equiv \mathbb{E}_t s_{t+1} - s_t + i_t^\star - i_t$ and falls with risk aversion ρ and the conditional variance σ_t^2. The two schedules differ only through the price-impact term $\partial s_t/\partial b_t^S$. The price impact lowers strategic investors' demand at every level of excess return and vanishes in the competitive limit. It also reduces the elasticity of investors' demand to changes in the interest rate differential. This term is an equilibrium object, which we derive below from the market-clearing condition in Equation 8.

Price Impact. Differentiating the market-clearing condition with respect to $b_t^{S,i}$—holding the other strategic investors' positions fixed and using the competitive schedule, $\partial b_t^C/\partial s_t = -1/(\rho\sigma_t^2)$—gives the price impact of a strategic investor i:

$$\frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i \rho \sigma_t^2}{B\rho\sigma_t^2 + (1-\lambda)} > 0, \tag{11}$$

which is increasing in the investor's own mass λ_i and decreasing in the size of the competitive fringe $1-\lambda$. Under symmetry ($\lambda_i = \lambda/N$), the individual price impact becomes $\frac{1}{N}\frac{\lambda\rho\sigma_t^2}{B\rho\sigma_t^2+(1-\lambda)}$.[10]

The structure of the market determines the magnitude of the price impact and, consequently, the relevance of strategic behavior. The price impact is negatively affected by the number of strategic traders, N, and positively related to the size of the strategic segment, λ. Therefore, it is larger in more concentrated markets characterized by a lower N and/or higher λ.

In our international portfolio model, strategic investors have a lower price impact on the equilibrium price of an asset compared to a closed-economy version. This is due to the presence of valuation effects on the supply of assets once denominated in domestic currency. By internalizing the effect that their demand has on the exchange rate, strategic investors also take into account how the value of the supply of foreign assets denominated in domestic currency varies when the exchange rate changes. This is reflected by the presence of B, the total supply of foreign assets, at the denominator of Equation (11).

Exchange Rate. Combining the demand schedules from Equation (10) with the market-clearing condition in Equation (8) and iterating forward yields an explicit expression for the exchange rate (up to a constant):

$$s_t = \mu \sum_{k=0}^{\infty} \mu^k \, \mathbb{E}_t \left(i^\star_{t+k} - i_{t+k} \right) + \frac{1-\mu}{b} \sum_{k=0}^{\infty} \mu^k \, \mathbb{E}_t \left(x_{t+k} \right), \tag{12}$$

where $b = \frac{B}{W}$ and $\mu \in (0,1)$ is a function of the market structure (λ, N) and the variance σ_t^2:

$$\mu = \frac{1 + B\rho\sigma_t^2 - \lambda\dfrac{N-1}{N} - \dfrac{\lambda^2}{N}}{\left(1 + B\rho\sigma_t^2\right)\left(1 + B\rho\sigma_t^2 - \lambda\dfrac{N-1}{N}\right) - \dfrac{\lambda^2}{N}}, \tag{13}$$

The exchange rate is a weighted average of current and expected future fundamentals ($i^\star_{t+k} - i_{t+k}$) and noise (x_{t+k}), where μ sets the weight on fundamentals relative to noise; it therefore measures how much information about fundamentals the exchange rate conveys. Its dependence on market structure is summarized by the following result.[11]

[10] In our analysis, we focus on the case of symmetric strategic investors due to the unavailability of comprehensive investor-level market share data. Importantly, all qualitative predictions are not altered by the symmetry assumption.

[11] The parameter μ relates to the magnification factor in Bacchetta and Van Wincoop (2006): in their work, information dispersion reduces the information content of exchange rates by amplifying the impact of noise traders. As in their paper, how μ responds to changes in λ is central to the amplification mechanism examined here.

Lemma 1 (Informativeness and market structure)**.** *Holding the conditional variance σ_t^2 fixed, μ is strictly decreasing in the size of the strategic segment λ and strictly increasing in the number of strategic investors N.*

By Lemma 1, greater market concentration—a larger strategic segment λ or fewer strategic traders N—lowers the informativeness of the exchange rate. Intuitively, a stronger price impact leads strategic investors to trade less aggressively on fundamentals, so the market-clearing exchange rate reflects noise-trader demand relatively more. In other words, When traders recognize that the residual supply curve is upward-sloped, quantities are restricted and less elastic, so prices become less informative. This aligns with the key intuition of Kyle (1989).

2.3 Theoretical Predictions

We illustrate the theoretical implications of strategic behavior for exchange rate dynamics. Specifically, we show that the presence of strategic investors may increase exchange rate volatility, exacerbate the disconnect between the exchange rate and underlying fundamentals, and lead to larger deviations from uncovered interest parity (UIP). We sketch the argument for each result here, while Appendix B.2 collects the detailed proofs.

Response to Shocks. In the model, an increase in the exchange rate arises due to either a positive noise shock or a negative fundamental shock. A positive noise shock, interpreted as either an increase in demand or a decrease in the supply of foreign assets, reduces the residual demand for these assets. This leads to a rise in the price of foreign assets and an appreciation of the exchange rate, despite no change in the underlying fundamentals. As the exchange rate increases, the excess return falls below its steady state, causing investors to reduce their foreign asset holdings and rebalance their portfolios in favor of domestic assets. Conversely, a negative fundamental shock, caused by contractionary monetary policy in the foreign country, reduces the interest rate differential, and directly increases the excess return on holding foreign assets. This is turn boosts investors' demand for foreign assets, leading to an appreciation of the foreign currency.

Proposition 1 (Response to Shocks)**.** *An increase in the size of the strategic segment λ amplifies the response of the exchange rate to noise shocks and dampens its response to fundamental shocks.*

To see this, consider the law motion of the exchange rate in Equation (12). Because the fundamental and noise processes are $AR(1)$ with persistence ρ_u and ρ_x, under rational expectations the conditional forecast k periods ahead simplifies to $\mathbb{E}_t(\Delta i_{t+k}) = \rho_u^k(i_t - i_t^\star) = \rho_u^k \Delta i_t$ and $\mathbb{E}_t(x_{t+k}) = \rho_x^k x_t$, with $\Delta i_t \equiv i_t - i_t^\star$. Substituting and summing the geometric series gives the closed-form solution for the exchange rate

$$s_t = -\frac{\mu}{1-\mu\rho_u}\Delta i_t + \frac{(1-\mu)}{(1-\mu\rho_x)b}x_t \tag{14}$$

Because Equation (14) holds in every period, the impulse response of the exchange rate $h \geq 0$ periods after a one-standard-deviation shock at t is

$$\mathrm{IRF}_s^{\Delta i}(h) = -\frac{\mu\,\rho_u^h}{1-\mu\rho_u}\sigma_u, \qquad \mathrm{IRF}_s^{x}(h) = \frac{(1-\mu)\,\rho_x^h}{(1-\mu\rho_x)\,b}\sigma_x, \tag{15}$$

where $\mathrm{IRF}_s^{\bullet}(h)$ denotes the deviation of the exchange-rate level s_{t+h} from its pre-shock path; at $h = 0$ these are the impact responses. Since μ is a decreasing (increasing) function of the size of the strategic segment λ (of the number of strategic investors N), the response to a fundamental shock is dampened while the response to a noise shock is amplified as λ increases (N decreases): at every horizon h, the magnitude of $|\mathrm{IRF}_s^{\Delta i}(h)|$ falls with λ and that of $|\mathrm{IRF}_s^{x}(h)|$ rises. Appendix Figure C.7 plots the impulse response functions for different values of λ.

Economically, this occurs because strategic investors, who internalize the negative impact their trades have on prices, consistently respond less aggressively to aggregate shocks, which affects the sensitivity of foreign asset demand. Therefore, in a world where investors behave strategically, the decline in demand for foreign assets in response to a noise shock is less pronounced compared to a competitive benchmark. The smaller decline in the demand for foreign bonds, due to strategic behavior, exerts additional upward pressure on the price of foreign bonds, thereby amplifying the effect of noise shocks on the exchange rate.

Conversely, in response to a fundamental shock, strategic investors increase their demand for foreign assets less than they would in a competitive market, again making total demand less sensitive to fundamental shocks. In this strategic environment, investors internalize their price impact, leading to a smaller increase in their holdings of foreign assets. As a result, the price of foreign assets rises less than it would in a competitive market, dampening the effect of the fundamental shock on the exchange rate.

Excess Return Predictability. An empirically robust evidence in exchange rate dynamics is the predictability of excess returns, commonly referred to as deviations from the Uncovered Interest Parity (UIP) (Fama, 1984). Our model predicts systematic deviations from UIP due to a non-zero net supply of foreign assets, regardless of the presence of strategic investors.

Proposition 2 (Excess Return Predictability)**.** *Excess returns are more predictable as the size of strategic investors in the market increases.*

Proposition 2 implies that strategic behavior in the FX market amplifies UIP deviations compared to a competitive market. To illustrate this, define the 1-period realized excess return $q_{t+1} = s_{t+1} - s_t - (i_t - i_t^*)$, and consider the corresponding one-period Fama regression:

$$q_{t+1} = \alpha + \beta^{\text{Fama}}(i_t - i_t^\star) + \nu_{t+1}, \tag{16}$$

where the coefficient, β^{Fama}, reflects deviations from the UIP theory. If UIP holds, the coefficient of the Fama regression is expected to be zero. However, in empirical studies, the Fama coefficient is systematically different from zero and negative. Based on Equation (16), the coefficient β^{Fama} is:

$$\beta^{\text{Fama}} \equiv \frac{\text{Cov}(q_{t+1},\, i_t - i_t^*)}{\text{Var}(i_t - i_t^*)} = -(1-\mu)\,\frac{1}{1-\mu\rho_u} \in (-1,\, 0), \tag{17}$$

which is negative, in line with empirical findings, and decreases as strategic behavior increases. In other words, when there are strategic investors, changes in the interest rate differential have a stronger impact on the excess return, and consequently on the risk premium.

In our framework, this happens because, as strategic investors' positions become less elastic to aggregate shocks, the market requires larger movements in the risk premium to induce investors to expand their positions. An increase in the size of strategic investors, λ, reduces market risk appetite, further necessitating a higher risk premium to absorb the net supply of foreign assets. This makes the excess return more responsive to changes in fundamentals, increasing its predictability. Formally, the expected excess return can be written as

$$\mathbb{E}_t\, q_{t+1} = \frac{1-\mu}{\mu}\, \frac{B(1+s_t) - X_t}{B} \tag{18}$$

The right-hand side represents the deviation from UIP, which is the risk premium required by investors for holding a foreign asset. The risk premium depends on two components: the net supply of foreign assets and the size of non-competitive investors, captured by $\frac{1-\mu}{\mu}$, which increases with the size of strategic investors, λ. The first component drives the deviation from UIP in the model, while the second component, $\frac{1-\mu}{\mu}$, amplifies the UIP deviation.[12]

Exchange Rate Excess Volatility. There is extensive evidence showing that exchange rates exhibit higher volatility compared to fundamentals, which is commonly referred to as the"excess volatility puzzle" (Obstfeld and Rogoff, 2000; Engel and Zhu, 2019). This puzzle provides evidence

[12] Appendix B.2 generalizes the result in Proposition 2 showing that the Fama coefficient is monotonically increasing in the time horizon k, and approaching zero for $k = \infty$. Therefore, our model does not explain the predictability reversal puzzle documented in Bacchetta and Van Wincoop (2010) and Engel (2016), which shows that expected excess returns reverse sign at longer horizons. This limitation is anticipated, as our focus is on an amplification mechanism without portfolio adjustment frictions (Bacchetta and Van Wincoop, 2010, 2019).

of the significant role of non-fundamental components in explaining exchange rate dynamics in the data.

Proposition 3 (Unconditional Variance)**.** *The unconditional variance of the exchange rate is*

$$\mathrm{Var}(s_t) = \frac{\mu^2}{(1-\mu\rho_u)^2}\,\frac{\sigma_u^2}{1-\rho_u^2} + \frac{(1-\mu)^2}{(1-\mu\rho_x)^2\, b^2}\,\frac{\sigma_x^2}{1-\rho_x^2}. \tag{19}$$

The result follows directly from Equation 14. Because s_t is a linear combination of two uncorrelated $AR(1)$ processes, Δi_t and x_t, with $\mathrm{Var}(\Delta i_t) = \sigma_u^2/(1-\rho_u^2)$ and $\mathrm{Var}(x_t) = \sigma_x^2/(1-\rho_x^2)$, there is no covariance term, and $\mathrm{Var}(s_t)$ is the sum of the variances of the two terms.

Whether a larger strategic segment raises or lowers exchange-rate volatility depends on how important noise shocks are relative to fundamentals. We summarize it in the following result.

Corollary 1 (Exchange Rate Excess Volatility)**.** *An increase in the size of strategic investors makes the exchange rate more volatile relative to fundamentals if and only if the noise-to-fundamental variance ratio exceeds the threshold* $\bar{\Omega}$.

$$\frac{\mathrm{Var}(x_t)}{\mathrm{Var}(i_t - i_t^\star)} \;>\; \overline{\Omega}, \qquad \overline{\Omega} \;\equiv\; b^2\, \frac{\mu\,(1-\mu\rho_x)^3}{(1-\mu)\,(1-\rho_x)\,(1-\mu\rho_u)^3}. \tag{20}$$

Following Engel and Zhu (2019), we measure excess volatility as the ratio of exchange-rate volatility (defined in Proposition 3) to fundamental volatility, $\sigma_u/\sqrt{1-\rho_u^2}$. Because fundamental volatility is independent of market structure, this ratio varies only with exchange-rate volatility; strategic investors therefore raise excess volatility exactly when they raise $\mathrm{Var}(s_t)$. By Corollary 1, this happens when noise volatility is sufficiently large relative to fundamentals.

Intuitively, this occurs because the presence of strategic investors reduce the informativeness of the exchange rate, placing relatively more weight on its noise component. A larger λ lowers μ (Lemma 1), which amplifies the exchange rate's response to noise shocks and dampens its response to fundamental shocks (Proposition 1). The net effect raises volatility whenever noise is the dominant driver—that is, when the variance ratio clears $\overline{\Omega}$ (Corollary 1). The quantitative assessment in Section 3 shows that this condition holds for any reasonable calibration of the fundamental and noise processes, although the effect of strategic behavior need not be monotonic from a theoretical perspective.

Exchange Rate Disconnect. One of the most well-established empirical findings in exchange rate dynamics is the disconnect between exchange rates and fundamentals (Meese and Rogoff, 1983; Cheung et al., 2005; Rossi, 2013). Specifically, variations in the interest rate differential explain only a small portion of the changes in exchange rates.

To measure the disconnect, we assess the explanatory power of the regression

$$s_{t+1} - s_t = \alpha + \beta \left(i_t - i_t^\star \right) + \nu_{t+1}, \tag{21}$$

in which the interest differential $i_t - i_t^\star$ is the fundamental driver of the one-period exchange-rate change $s_{t+1} - s_t$.

Proposition 4 (Exchange Rate Disconnect)**.** *The share of the variance of exchange-rate changes explained by fundamentals in regression (21) is*

$$R^2 = \frac{\mathrm{Var}(i_t - i_t^\star)}{\mathrm{Var}(\Delta s_{t+1})} \left[1 + \beta^{Fama} \right]^2. \tag{22}$$

The R^2 depends on two factors: the ratio of the fundamental variance to the variance of exchange-rate changes, and the Fama coefficient from Proposition 2. An increase in the size of strategic investors has an ambiguous effect on the first factor, because strategic behavior moves the volatility of exchange-rate changes $\mathrm{Var}(\Delta s_{t+1})$ in a direction that depends on the relative variance of the two shocks. By contrast, it makes the exchange rate more predictable, i.e. β^{Fama} more negative (Proposition 2), which reduces the second factor. Although the first factor is not signed, their combined effect is unambiguous, as the following result establishes.

Corollary 2 (Disconnect and Strategic Size)**.** *R^2 is strictly decreasing in the size of the strategic segment λ, for any stationary parametrization with $\sigma_x > 0$; that is, the disconnect between the exchange rate and fundamentals increases in the size of strategic investors.*

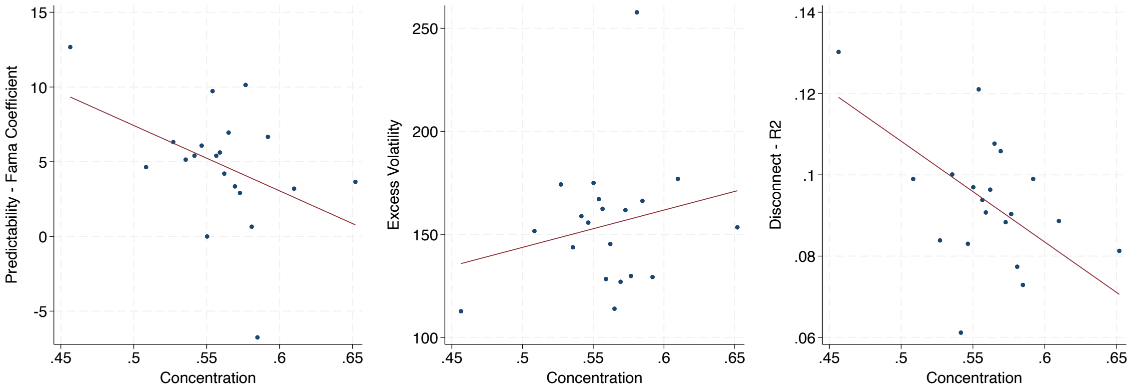
This follows because strategic investors reduce the informativeness of the exchange rate: by shifting weight toward non-fundamental noise, they leave fundamentals accounting for a smaller share of exchange-rate movements.

2.4 Testing Predictions

We leverage the heterogeneity in market concentration across currencies to test the implications of our theory. The model delivers three distinct testable relationships between exchange rate dynamics and the level of strategic behavior: (i) currencies with a larger presence of strategic investors are more predictable; (ii) higher levels of strategic behavior result in greater excess volatility in exchange rates; and (iii) the disconnect between exchange rates and fundamentals increases with the level of strategic behavior.

We test the model's predictions using a set of 10 currencies merged with U.S. CFTC transaction

Figure 2: Testing Model Predictions



Notes: The figure presents binscatter plots showing the positive relationship between the level of strategic behavior and the excess return predictability (left panel), the excess volatility (central panel), and the disconnect (right panel) of the exchange rate in the actual data. Concentration is the share of open interest held by the top eight traders in the future FX market. Data is from the U.S. CFTC spanning from 2006 to 2016. Exchange rate predictability is represented by the regression coefficient from the Fama regression in Equation (16). The exchange rate disconnect is measured using the R^2 from the regression in Equation (21), while excess volatility is calculated as the ratio of exchange rate volatility to the volatility of the interest rate differential. To measure predictability, excess volatility, and disconnect, we use 2-year rolling window regressions based on monthly average exchange rate data. The resulting data are demeaned at the currency and year levels, and the Fama coefficient, excess volatility ratio, and disconnect values are winsorized at the 1% level. The South African Rand is excluded from the set of 11 currencies. The estimated coefficients are reported in Table 1, while Appendix A provides further details on the data used.

data, available from June 2006 to December 2016.[13] An advantage of these data is that they cover both major and non-major currencies and provide standardized measures of trading activity over a relatively long period. However, the data also have important limitations: they cover only a subset of market participants and are collected from the U.S. futures market, rather than including other major OTC trading centers. As a result, our measure of trading concentration provides an informative but necessarily incomplete representation of concentration in global foreign exchange markets.

We use the concentration ratio of the top eight investors, as reported by the U.S. CFTC, as our proxy of strategic behavior in the foreign exchange market (λ). The assumption is that variations in concentration ratios between currencies may reflect differences in the presence and influence of traders who engage in strategic trading. We correlate this information with time-varying metrics of exchange rate disconnect, predictability and excess volatility. We utilize a 2-year rolling window with monthly exchange rate data to create time-varying indexes for exchange rate predictability,

[13]We exclude the South African Rand from our analysis due to a limited number of time observations in the CFTC dataset. To reduce noise in weekly transactions, we aggregate the data to a monthly level.

Table 1: Testing Model Predictions

	(1) Fama Coefficient	(2) Excess Volatility	(3) Disconnect - R2
λ	-43.759***	180.925***	-0.248***
	(8.288)	(69.343)	(0.068)
Constant	29.304***	53.229	0.232***
	(4.684)	(38.980)	(0.038)
Currency & Year FEs	Yes	Yes	Yes
Observations	900	900	900

Notes: The table reports the relationship between λ and the variables of interest. λ represents the net concentration ratio by the top eight reporting traders operating in the future FX market. Variable of interest are: Fama coefficient (Column (1)), exchange rate excess volatility (Columns (2)); exchange rate disconnect/R^2 (Column (3)). The exchange rate disconnect is measured using the Adjusted R^2 from the regression in Equation (21), while excess volatility is calculated as the ratio of exchange rate volatility to the volatility of the interest rate differential. λ is measured monthly from 2006 to 2016 using the U.S. CFTC data. To measure excess volatility and disconnect, we use a two-year periods rolling window and exchange rate data at monthly level. Values of the excess volatility ratio and disconnect are winsorized at 1%. All regressions include currency and year fixed effects. Standard errors in parenthesis are clustered at the currency level. Significance level:* $p<0.10$, ** $p<0.05$, *** $p<0.01$. Currencies considered are: Euro, Japanese Yen, Brazilian Real, Canadian Dollar, Swiss Franc, Australian Dollar, Mexican Peso, British Pound, Russian Ruble, and New Zeland Dollar. Appendix A provides additional information on the data used.

disconnect and excess volatility. We measure exchange rate predictability as the coefficient on the interest rate differential from the Fama regression in Equation (16), exchange rate disconnect using the R^2 from the regression in Equation (21), and excess volatility as the ratio of exchange rate volatility to the volatility of the interest rate differential. The panel nature of our dataset enable us to incorporate currency and year fixed effects, mitigating potential concerns regarding spurious correlation and strengthening the validity of the empirical evidence.[14]

Figure 2 presents binscatter plots showing empirical relationships consistent with the predictions of our theoretical framework. The left panel shows that, on average, currencies with more concentrated trading flows, held by fewer investors, tend to be more predictable. The middle panel documents a strong, positive, and statistically significant relationship between our measure of strategic behavior in exchange rate markets and the excess volatility of exchange rates. Finally, the right panel reveals that as the presence of strategic investors in the market increases, currencies become more disconnected from fundamentals, as indicated by the decreasing estimated R^2.

Table 1 reports the estimated coefficients along with the corresponding standard errors, clustered at the country level. We find that a currency traded in a market with a 10% higher concentration ratio exhibits an excess volatility ratio that is approximately 12% higher than the sample

[14] The results remain unchanged when we increase the rolling window size to 3 and 4 years, and using different proxies for strategic behavior, such as the number of active traders.

average. Similarly, a currency traded in a market with a 10% higher concentration ratio exhibits 18% lower predictive power compared to the average R^2 observed in the sample.

3 Quantitative Assessment

To what extent can heterogeneity in price impact account for exchange rate puzzles? We approach this question through the lens of the framework developed in Section 2. While its analytical tractability affords sharp comparative statics linking the size of the strategic segment to the manifestation of these puzzles, we are mindful that such parsimony is not without cost. The model's deliberately stylized structure necessarily circumscribes its scope, and thus the ensuing exercise is best interpreted as a disciplined theoretical benchmark rather than a definitive quantification of the mechanism's empirical relevance.

Moreover, a precise quantitative answer would require transaction-level data and information on the identity and positioning of individual players in the foreign exchange market, which is not available to us. In particular, two parameters are difficult to discipline directly from the data: the size of the strategic segment λ and the coefficient of relative risk aversion ρ. Concentration ratios provide a natural empirical proxy for λ but are an imperfect measure of strategic behavior — large investors may refrain from strategic action because of regulatory scrutiny, while smaller but sophisticated traders may behave strategically despite a smaller market share. Risk aversion, in turn, is unobservable and the literature offers a wide range of values. Rather than committing to a single point estimate, we adopt a benchmark calibration disciplined by CFTC concentration data and existing studies, and then vary λ and ρ across plausible ranges to map out the contribution of the mechanism under alternative assumptions. This approach reflects the inherent uncertainty about the true level of strategic behavior in foreign exchange markets and provides a range of estimates rather than a single number.

3.1 Range of Quantitative Estimates

We calibrate our framework to a panel of 18 currencies, run counterfactuals against a competitive benchmark, and quantify the contribution of the micro-mechanism to the exchange rate disconnect, the forward premium puzzle, and exchange rate volatility through the lens of a monetary macro model.

Calibration. The benchmark parametrization is reported in Table 2. We calibrate the model using monthly exchange rate data on 18 currencies against the U.S. Dollar from 1993 to 2019.[15]

[15]The currencies are listed in Appendix A.

Without loss of generality we set $\bar{r} = 0$, which implies that the interest rate differential reduces to $i_t - i_t^\star = u_t$. We assume that covered interest rate parity holds and compute the one-month interest rate differential as the difference between the log one-month forward and the log spot exchange rate. We model the fundamental as an AR(1) process and estimate its volatility and persistence currency by currency, calibrating $\sigma_u = 0.005$ and $\rho_u = 0.85$ to match the cross-currency averages. The variance of the one-period exchange rate change, σ_t, is calibrated to its empirical counterpart, 0.028.

The noise process x_t is calibrated to match exchange rate dynamics. We set $\rho_x = 0.9$, high enough to keep the exchange rate sufficiently close to a random walk, and choose σ_x to match the empirical volatility of the one-period exchange rate change conditional on the calibrated market-structure parameters. As Equation (12) makes clear, exchange rate dynamics depend jointly on the noise process and on the market structure, so the noise volatility required to match a given empirical target depends on λ and N. We therefore first calibrate λ and N, and then estimate σ_x residually.

We use data from the U.S. Commodity Futures Trading Commission to calibrate the market-structure parameters. We set $N = 8$ and $\lambda = 0.572$, the latter being the average concentration ratio of the top eight reportable traders in the FX futures market over 2006–2016. Given these values, we obtain $\sigma_x = 0.132$.[16] We follow Bacchetta and Van Wincoop (2019) and set $b = 0.33$ — implying that foreign assets account for roughly one third of domestic financial wealth — and the relative risk aversion coefficient at $\rho = 50$.

Results. We assess the impact of strategic behavior by comparing simulated moments under the benchmark parametrization to those generated by an otherwise identical competitive economy in which $\lambda = 0$ (all investors are atomistic price takers). For each model specification we run 1,000 simulations of 6,000 periods with a 3,000-period burn-in, and report the moments summarizing the exchange rate response to fundamental and noise shocks, the Fama coefficient, exchange rate volatility, and the disconnect R^2. Table 3 reports the results under different parametrization in percentage deviation from a competitive model, while point estimates are reported in the Appendix Table C.7.[17]

Under the benchmark, strategic behavior amplifies the response to noise shocks more than it

[16]Appendix Figure B.6 shows that the calibrated value of σ_x exceeds the minimum volatility of the noise process required for exchange rate volatility to increase monotonically with the degree of strategic behavior, as characterized in Proposition 3. Moreover, Appendix Figure C.8 shows that the implied σ_x, for different values of λ and N and given the targeted dynamics of the exchange rate, consistently remains above the minimum value reported in Appendix Figure B.6. This confirms the quantitative relevance of the conditions underlying Proposition 3 over the range of parameter values considered.

[17]Appendix Table C.7 also reports the corresponding moments from the data, highlighting that, despite the parsimonious framework, the model is able to broadly match the targeted moments.

Table 2: Benchmark Parametrization

Parameters	Value	Target
ρ	50	Bacchetta and Van Wincoop (2019)
b	0.333	Home Bias
ρ_u	0.85	Average persistence AR(1) Δi_t
σ_u	0.005	Average StD innovation AR(1) Δi_t
σ_t	0.028	σ_t (Volatility ER change)
ρ_x	0.9	ER Random Walk/Average Disconnect
λ	0.572	Net concentration ratio (Top 8) – US CFTC
N	8	Number of traders related to λ – US CFTC
σ_x	0.132	Average StD ER change

Notes: The table summarizes the parametrization used in the basic framework. For each parameters, we report the value used in the model, the corresponding moment and data used to calibrate, and, if applicable, the target moment used to estimate it. Appendix A provides additional information on the data used.

dampens the response to fundamental shocks. A one-standard-deviation noise shock raises the exchange rate by 14.66% in the competitive economy and by 15.51% in the strategic benchmark, a 5.8% increase.[18] The corresponding dampening of the fundamental response is an order of magnitude smaller (approximately 2%). The asymmetry reflects two features of the calibration: the noise process is more persistent than the fundamental ($\rho_x = 0.9$ vs. $\rho_u = 0.85$), and the noise component enters with a coefficient scaled by $1/b$, which amplifies its weight given the calibrated home bias. The presence of strategic investors moves the Fama coefficient from -0.204 to -0.217 (roughly 6% more negative), increases exchange rate volatility by approximately 6%, and reduces the disconnect R^2 from 0.083 to 0.073, a 12% decrease. Under this calibration, the mechanism contributes meaningfully to the exchange rate disconnect and modestly to the forward premium and excess volatility puzzles.

The benchmark sets λ at the cross-currency average. The contribution of the mechanism scales nontrivially with this parameter. Reducing the average market share of a strategic investor from $\lambda/N \approx 7\%$ to 5% shrinks the noise amplification from 5.8% to 2.1% and the disconnect contribution from 12% to 4.7%. Conversely, raising the average market share to 9% — still well within empirical concentration ranges for non-major USD pairs (Figure 1) — amplifies the noise response by 12.8%, makes the Fama coefficient 14.4% more negative, raises volatility by 12.3%, and accounts for a 24% reduction in the disconnect R^2.

Given the uncertainty around the calibration of λ due to data limitation, Appendix Figure

[18] Appendix Figure C.7 shows the impulse-response functions of the key variable of interests to both fundamental and noise shocks as described in Proposition 1 given the calibration in Table 2.

Table 3: Summary of the Quantitative Results

Moments from Simulated Data (% Deviation from Competitive Model)	(1) Response to Shock Noise (Fundamental)	(2) Fama Coefficient	(3) Exchange Rate Volatility (St.D.)	(4) Exchange Rate Disconnect
Benchmark Case	0.058 (0.017)	-0.064	0.055	-0.122
Size Strategic Investors				
Low ($\lambda/N = 5\%$)	0.021 (0.006)	-0.024	0.02	-0.047
High ($\lambda/N = 9\%$)	0.128 (0.038)	-0.144	0.123	-0.247
Risk Adversion				
Low ($\rho = 30$)	0.068 (0.011)	-0.073	0.061	-0.117
High ($\rho = 70$)	0.05 (0.021)	-0.057	0.049	-0.118
Periods per Simulation	6,000	6,000	6,000	6,000
Burn-in Periods	3,000	3,000	3,000	3,000
Number of Simulations	1,000	1,000	1,000	1,000

Notes: The Table reports the impact of strategic investors on exchange rate dynamics as percentage deviations from the competitive model using simulated data (e.g., 0.01 unit indicates a 1% deviation from the competitive model). For comparison, the first row presents the impact of strategic investors on exchange rate moments in the benchmark case. Each column reports percentage deviations from the competitive model. Column (1) shows the amplification or dampening of the exchange rate response to a one-standard-deviation noise and fundamental shock. Column (2) reports the percentage deviation in the Fama regression beta. Column (3) reports the percentage deviation in exchange rate volatility, measured by the standard deviation, and Column (4) provides the percentage deviation in the estimated R^2 from the disconnect regression. The results are obtained from 1,000 simulations, with each iteration running for 6,000 periods and a burn-in of 3,000 periods. All the other parameters are held constant across scenarios; see Table 2 for details.

C.9 plots the moments across the full range of λ. The response to noise shocks rises from approximately 0% in the competitive limit to nearly 38% as $\lambda \to 1$, and the disconnect contribution increases monotonically over the same range. The mechanism's contribution to exchange rate puzzles therefore depends critically on the assumed size of the strategic segment, and is substantially larger at the concentration levels observed for non-major currencies than at the all-currency average used in the benchmark. Thus, while the model is stylized and its calibration is subject to uncertainty, the results indicate that the mechanism can have economically meaningful bite for sufficiently high values of λ.

Finally, lowering risk aversion to $\rho = 30$ raises the noise response by 6.8% relative to the competitive model and increases volatility by 6.1%; raising risk aversion to $\rho = 70$ moderates both effects. The intuition is that risk-tolerant strategic investors are willing to take larger positions, which makes their internalized price impact more consequential for equilibrium prices.[19]

[19]This pattern operates through the equilibrium informativeness parameter μ rather than through the standard risk-premium channel emphasized in Itskhoki and Mukhin (2021), where higher risk aversion makes excess returns more sensitive to flows. In our setup, higher ρ lower μ, but it also raises $B\rho\mathrm{Var}_t(s_{t+1})$ in both the strategic and competitive benchmarks, so the differential impact of strategic behavior on dynamics is

3.2 Taking Stock

The quantitative exercise yields two main findings. First, at the average level of strategic behavior observed in our CFTC sample, the mechanism accounts for a modest portion of the exchange rate disconnect and only a limited share of the forward premium and excess volatility puzzles. While these effects are non-negligible, we do not view them as constituting a resolution of any of the puzzles. Second, the contribution scales nonlinearly with the size of the strategic segment. At concentration levels observed for non-major USD currency pairs, the mechanism becomes quantitatively significant—particularly for the disconnect—highlighting the importance of accurately measuring market concentration.

We emphasize, however, that this exercise is subject to important limitations. The CFTC data pertain to the U.S. FX futures market and may not provide a comprehensive measure of concentration in the broader FX market. Moreover, while the model delivers sharp predictions, it abstracts from several features that may be quantitatively relevant, such as intermediary frictions that could interact with strategic behavior in important ways (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021).

4 Conclusions

Concentration in the foreign exchange rate markets and the implied heterogeneity in price impact may play an important role in understanding exchange rate dynamics.

In this paper, we explore the implication of strategic behavior in a monetary model of exchange rate determination. We show that strategic behavior reduces the informativeness of the exchange rate by amplifying the response to non-fundamental shocks while dampening the response to fundamental shocks. As a result, heterogeneity in price impact helps to explain the weak empirical link between fundamentals and exchange rates, as well as the excess volatility observed in exchange rate movements. Although our model is stylized to derive fundamental insights and analytic results, we provide empirical evidence supporting the theoretical predictions using a panel of 10 currencies.

This paper represents a step forward in incorporating microstructure institutions in the analysis of exchange rate dynamics. Our framework is tractable and can be integrated into macro models of exchange rate determination. As shown in previous literature, the introduction of investor heterogeneity qualitatively and quantitatively alters conclusions regarding optimal monetary and exchange rate policies. It also calls for additional efforts in documenting and studying investors' heterogeneity in foreign exchange rate markets.

muted. The two channels are not in conflict; they act on different objects.

References

BACCHETTA, P. AND E. VAN WINCOOP (2006): "Can information heterogeneity explain the exchange rate determination puzzle?" *American Economic Review*, 96, 552–576.

——— (2010): "Infrequent portfolio decisions: A solution to the forward discount puzzle," *American Economic Review*, 100, 870–904.

——— (2019): "Puzzling exchange rate dynamics and delayed portfolio adjustment," Tech. rep., National Bureau of Economic Research.

CANDIAN, G. AND P. DE LEO (2022): "Imperfect exchange rate expectations," *Available at SSRN 3929756*.

CHEUNG, Y.-W., M. D. CHINN, AND A. G. PASCUAL (2005): "Empirical exchange rate models of the nineties: Are any fit to survive?" *Journal of international money and finance*, 24, 1150–1175.

COCHRANE, J. (2015): "The new WM/Reuters fixing methodology doesn't fix everything," *ITG Thought Leadership*.

CORSETTI, G., A. DASGUPTA, S. MORRIS, AND H. S. SHIN (2004): "Does one Soros make a difference? A theory of currency crises with large and small traders," *The Review of Economic Studies*, 71, 87–113.

CORSETTI, G., P. PESENTI, AND N. ROUBINI (2002): "The role of large players in currency crises," in *Preventing Currency Crises in Emerging Markets*, University of Chicago Press, 197–268.

ENGEL, C. (2016): "Exchange rates, interest rates, and the risk premium," *American Economic Review*, 106, 436–74.

ENGEL, C. AND K. D. WEST (2005): "Exchange rates and fundamentals," *Journal of political Economy*, 113, 485–517.

ENGEL, C. M. AND F. ZHU (2019): "Exchange rate puzzles: evidence from rigidly fixed nominal exchange rate systems," *BIS Working Paper*.

EVANS, M. D. AND R. K. LYONS (2002): "Order flow and exchange rate dynamics," *Journal of political economy*, 110, 170–180.

——— (2006): "Understanding order flow," *International Journal of Finance & Economics*, 11, 3–23.

Fama, E. F. (1984): "Forward and spot exchange rates," *Journal of monetary economics*, 14, 319–338.

Gabaix, X. and M. Maggiori (2015): "International liquidity and exchange rate dynamics," *The Quarterly Journal of Economics*, 130, 1369–1420.

He, Z. and A. Krishnamurthy (2013): "Intermediary asset pricing," *American Economic Review*, 103, 732–770.

Itskhoki, O. and D. Mukhin (2021): "Exchange rate disconnect in general equilibrium," *Journal of Political Economy*, 129, 2183–2232.

Jeanne, O. and A. K. Rose (2002): "Noise trading and exchange rate regimes," *The Quarterly Journal of Economics*, 117, 537–569.

Kacperczyk, M., J. Nosal, and S. Sundaresan (2025): "Market power and price informativeness," *Review of Economic Studies*, 92, 1955–1986.

Kyle, A. S. (1985): "Continuous auctions and insider trading," *Econometrica*, 1315–1335.

——— (1989): "Informed speculation with imperfect competition," *The Review of Economic Studies*, 56, 317–355.

Lyons, R. K. et al. (2001): *The microstructure approach to exchange rates*, vol. 333, Citeseer.

Meese, R. A. and K. Rogoff (1983): "Empirical exchange rate models of the seventies: Do they fit out of sample?" *Journal of international economics*, 14, 3–24.

Mussa, M. (1982): "A model of exchange rate dynamics," *Journal of political economy*, 90, 74–104.

Nimark, K. (2017): "Dynamic higher order expectations," *CEPR Discussion Paper No. DP11863.*

Obstfeld, M. and K. Rogoff (2000): "The six major puzzles in international macroeconomics: is there a common cause?" *NBER macroeconomics annual*, 15, 339–390.

Osler, C. (2014): "The Fix is In: how banks allegedly rigged the US $5.3 trillion foreign exchange market," *The Conversation.*

Osler, C., A. Turnbull, et al. (2016): "Dealer trading at the fix," Tech. rep.

Rossi, B. (2013): "Exchange rate predictability," *Journal of economic literature*, 51, 1063–1119.

Stavrakeva, V. and J. Tang (2020): "Deviations from fire and exchange rates: A GE theory of supply and demand," *working paper.*

Appendix

A Empirics

A.1 Data

We use three main sources of data:

- We use data on 18 currencies from December 1993 to December 2019. The currencies considered are: Euro, Japanese Yen, Argentinian Peso, Brazilian Real, Canadian Dollar, Swiss Franc, Australian Dollar, Chilean Peso, Indian Rupee, Mexican Peso, British Pound, South African Rand, Russian Ruble, Swedish Krona, Turkish Lira, New Zealand Dollar, Singapore Dollar, Norwegian Krone. The panel is not balanced.

  We obtain data for the spot and one-month forward exchange rates at a daily frequency from Datastream and Thompson Reuters. All exchange rates are defined against the US Dollar. To calculate the one-month interest rate, we took the difference between the logarithm of the one-month forward exchange rate and the logarithm of the spot exchange rate. We then computed monthly averages for the spot exchange rates and the one-month interest rate differentials.

- We use data from the U.S. Commodity Futures Trading Commission (CFTC) on investors' financial positions, among which currency positions: https://www.cftc.gov/MarketReports/CommitmentsofTraders/HistoricalCompressed/index.htm. The U.S. Commodity Futures Trading Commission (CFTC) data provides detailed information on several aspects within the currency futures market, including net open interest positions held by asset managers, institutional investors, and leveraged funds, as well as measures of concentration and the number of reportable traders. Data is reported on a weekly basis and spans the years 2006 to 2016 for 11 currency pairs. These pairs include both major and non-major USD currency pairs, reflecting the diversity of assets traded within the currency futures market. Major currency pairs typically involve the U.S. dollar and another major global currency, such as the Euro or Japanese Yen, while non-major pairs may involve currencies from emerging markets or smaller economies. Table A.4 in Appendix A reports summary statistics on key variables. Figures A.3 and A.4 in Appendix A respectively show the concentration ratio, and the number of reportable traders in the future FX market per currency and trader group.

- We use data on foreign exchange market concentration from the New York Fed FXC reports, which are available online here https://www.newyorkfed.org/fxc/fx-volume-survey.

The data covers the same set of currencies from 2005 to 2020. This panel is also unbalanced. The FXC survey is conducted biannually, specifically in April and October. We also use information on the geographical distribution of OTC foreign exchange turnover from the Triennial Central Bank Survey of Foreign Exchange and Derivatives Market Activity. The 2019 edition can be found here (Table 6): https://www.bis.org/publications/201909-commentary-otc-derivatives.pdf. The same survey provides data on the number of banks accounting for 75 percent or more of the FX turnover. For instance, for the year 2007, information can be found on Table B.4 here: https://www.bis.org/publications/triennial-central-bank-survey-foreign-exchange-and-derivatives-market-activity-2007-final-results.pdf

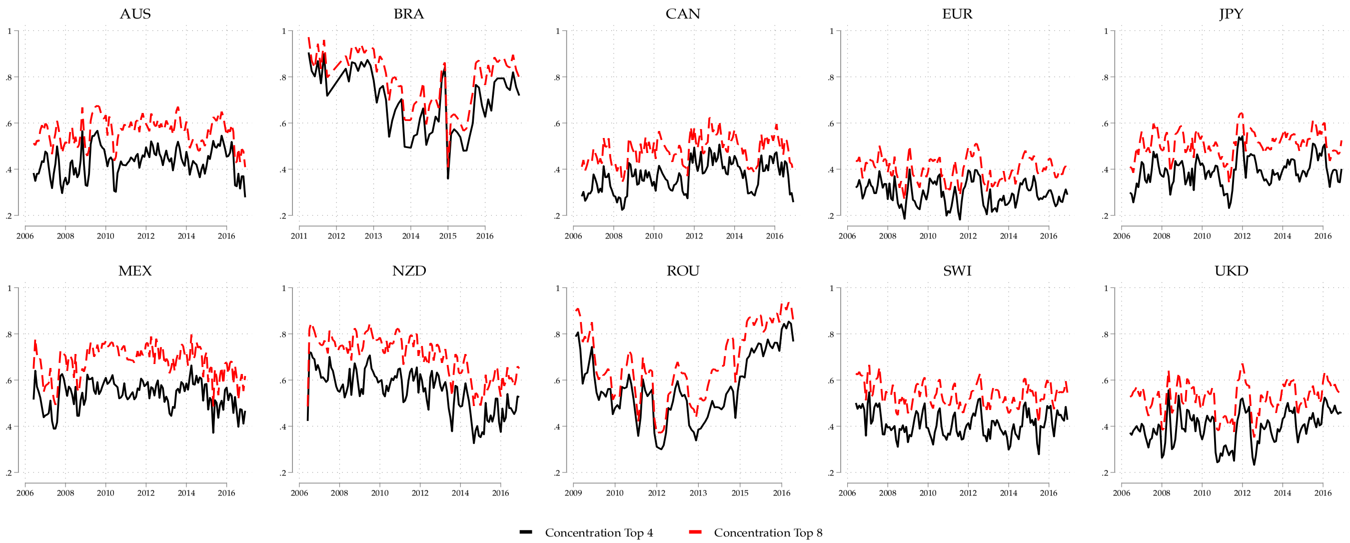
A.2 Additional Figures and Tables

Table A.4: Summary Statistics - Key Variables from CFTC dataset

	Currency										
	AUS	BRA	CAN	EUR	JPY	MEX	NZD	ROU	SWI	UKD	Total
Imbalances (Mil USD): Intermediaries	-16.180	-6.440	-12.658	28.322	18.114	-32.066	-6.517	-7.947	5.379	22.880	-0.136
Institutional Investors	-7.799	2.060	2.166	-5.080	11.954	17.415	-1.776	1.024	-0.632	-22.889	-0.544
Hedge Funds	24.303	3.980	-1.949	-26.026	-20.759	15.297	7.938	5.004	-3.839	10.270	1.152
Others	-4.234	0.303	5.850	9.987	2.196	-2.221	-0.283	0.969	0.263	-7.536	0.530
Concentration: Top 4 (Net)	0.433	0.699	0.368	0.295	0.392	0.537	0.553	0.566	0.411	0.396	0.448
Top 8 (Net)	0.564	0.795	0.484	0.399	0.513	0.680	0.704	0.676	0.537	0.523	0.572
Number: Intermediaries	7.788	4.693	7.448	13.411	8.921	7.075	6.764	6.327	6.180	7.272	7.830
Institutional Investors	6.115	0.084	5.428	15.684	7.773	6.200	2.744	0.000	1.974	7.333	6.442
Hedge Funds	15.614	6.129	13.881	25.069	19.697	16.122	9.867	4.883	8.898	16.649	14.738
Others	5.361	2.340	7.266	12.580	6.352	6.866	4.201	0.482	0.199	5.114	5.952

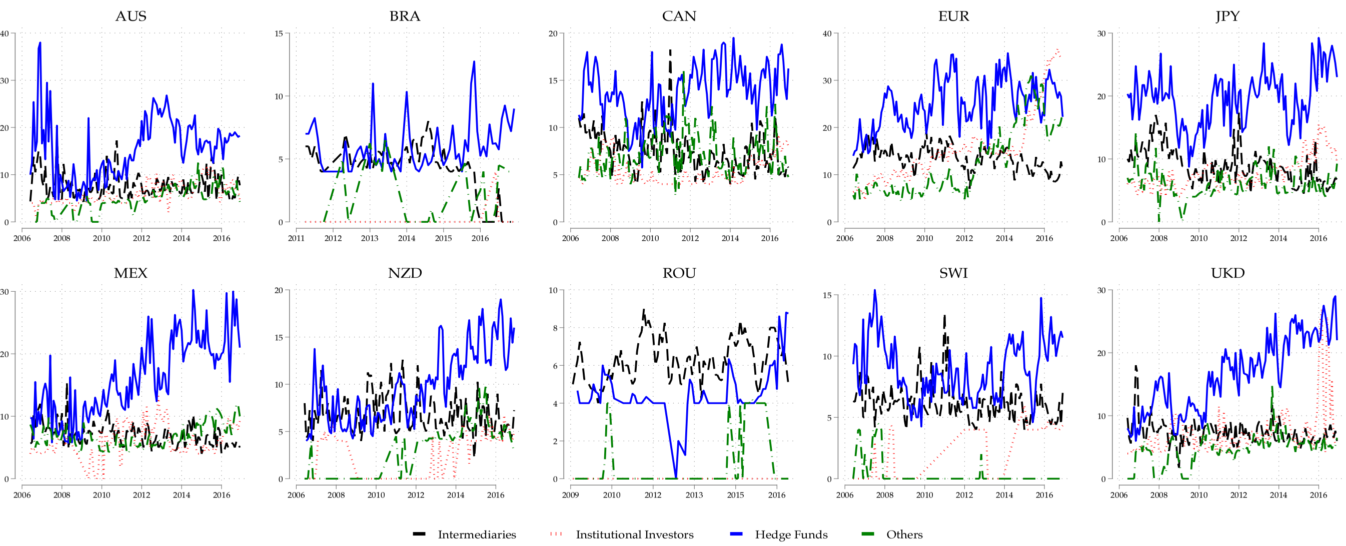
Notes: Report the means of the main variables in the CFTC dataset by currency pair. Net open interest positions are in millions of dollars ($), concentration ratios are expressed in percentages, and the number of traders is in count. The reported mean statistic is calculated based on a panel of weekly observations spanning from 2006 to 2018.

Figure A.3: Concentration Ratios by Currency Pairs - CFTC data



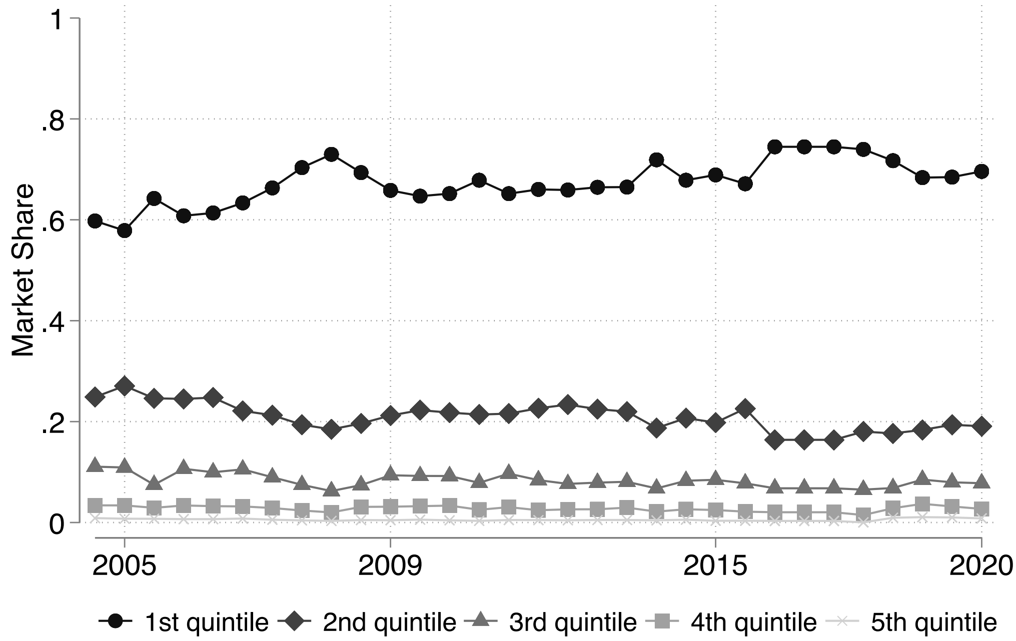
Notes: The figure shows the average percentages of open interest held, referred to as Concentration Ratios, by the largest four (black line) and eight (red line) reportable traders in the future FX market by currency pair. These concentration ratios are based on 'Net Position' and are calculated after offsetting each trader's equal long and short positions. The data is sourced from the U.S. Commodity Futures Trading Commission (CFTC) and spans from 2006 to 2016. Data are quarterly averages for each currency pair. Appendix A provides additional details regarding the data used.

Figure A.4: Number of Traders by Currency Pairs - CFTC data



Notes: The figure shows the numbers of reportable traders in the future FX market by currency pair. For each currency pair, we report the number of Dealers (black dashed line), Institutional Investors (red dotted line), Hedge Funds (blue line) and Other Reportable Traders (green dashed line). The data is sourced from the U.S. Commodity Futures Trading Commission (CFTC) and spans from 2006 to 2016. Data are quarterly averages for each currency pair. Appendix A provides additional details regarding the data used.

Figure A.5: Market Concentration – NY OTC Foreign Exchange Market



Notes: The figure shows the investors' market share in the New York foreign exchange market by quintile. Market share are computed in terms of total transactions across all currencies. Data are from the biannual NY Fed FXC report, spanning from 2005 to 2020.

Table A.5: Summary Statistics - Exchange Rate Concentration

Panel A. Geographic Concentration

Market	1995	1998	2001	2004	2007	2010	2013	2016	2019
Hong Kong	5,6	3,8	4,0	4,1	4,2	4,7	4,1	6,7	7,6
Japan	10,3	7,0	9,0	8,0	5,8	6,2	5,6	6,1	4,5
Singapore	6,6	6,9	6,1	5,1	5,6	5,3	5,7	7,9	7,7
Switzerland	5,4	4,4	4,5	3,3	5,9	4,9	3,2	2,4	3,3
UK	29,3	32,6	32,0	32,0	34,6	36,7	40,8	36,9	43,1
US	16,3	18,3	16,1	19,1	17,4	17,9	18,9	19,5	16,5
Others	26,6	27,1	28,2	28,4	26,3	24,2	21,7	20,4	17,2

Panel B. Within Market Concentration

Market	1995	1998	2001	2004	2007	2010
Hong Kong	22	26	14	11	12	16
Japan	24	19	17	11	9	8
Singapore	25	23	18	11	11	10
Switzerland	5	7	6	5	3	2
United Kingdom	20	24	17	16	12	9
United States	20	20	13	11	10	7
Others		37	32	25	28	27

Notes: The top panel of the table reports the percentage share of turnover intermediated in each foreign exchange rate market. The values are expressed in percentage terms. The source of the data is the BIS Triennial Survey, covering the period from 1995 to 2019. The bottom panel provides information on the number of investors accounting for 75 percent of the turnover in each foreign exchange rate market. The data is sourced from the BIS Triennial Survey, covering the period from 1995 to 2010.A.

Table A.6: Summary Statistics - Exchange Rates

Panel A. Monthly frequency

Variables	Mean	SD	Min	Max	N
Exchange rate return (in %)	0.334	0.548	-0.130	1.910	18
Exchange rate level volatility (StD)	0.344	0.353	0.115	1.367	18
Volatility of exchange rate changes (StD)	0.029	0.012	0.013	0.055	18
Fama coefficient (UIP)	-0.603	1.317	-2.688	2.357	18
R-Squared (Exchange rate disconnect)	0.030	0.065	0.000	0.265	18

Panel B. Weekly frequency

Variables	Mean	SD	Min	Max	N
Exchange rate return (in %)	0.077	0.127	-0.030	0.442	18
Exchange rate level volatility (StD)	0.342	0.350	0.115	1.351	18
Volatility of exchange rate changes (StD)	0.014	0.006	0.006	0.030	18
Fama coefficient (UIP)	-0.493	0.789	-1.479	1.194	18
R-Squared (Exchange rate disconnect)	0.037	0.064	0.000	0.261	18

Notes: The table presents summary statistics for various moments of exchange rates across different currencies. Panel A. reports statistics calculated at a monthly frequency, while Panel B. reports statistics calculated at a weekly frequency. We calculated the moments for each currency individually and then compiled a table to summarize the variation of these moments across currencies. The data cover the period from 1993 to 2019 for 18 currencies as described in Appendix A.

B Model Derivations and Proofs

This appendix derives the closed-form market equilibrium and collects the formal proofs of all propositions in Section 2.

B.1 Derivation of the Market Equilibrium

We obtain the equilibrium in three steps.

Step 1: Demand Schedules

At date t, investor j chooses foreign-bond holdings b_t^j to solve

$$\max_{b_t^j} \ \mathbb{E}_t^j(w_{t+1}^j|\Omega_t^j) - \frac{\rho}{2}\mathrm{Var}_t^j(w_{t+1}^j|\Omega_t^j)$$

$$\text{s.t.} \quad w_{t+1}^j = (\omega - b_t^j)i_t + (i_t^\star + s_{t+1} - s_t)b_t^j$$

Because investors have symmetric rational-expectations information sets, $\Omega_t^j = \Omega_t$, and we drop the j superscript on the conditional moments. The investor splits the endowment ω between the domestic bond (holding $\omega - b_t^j$, gross return i_t) and the foreign bond (holding b_t^j, gross return $i_t^\star + s_{t+1} - s_t$). Collecting terms in b_t^j,

$$w_{t+1}^j = \omega\, i_t + b_t^j\,(i_t^\star + s_{t+1} - s_t - i_t)$$

It follows that the portfolio decision depends entirely through $b_t^j\ (i_t^\star + s_{t+1} - s_t - i_t)$. At date t the variables i_t, $i_t^\star$, s_t are known (they belong to Ω_t); the only random term is next period's exchange rate s_{t+1}. Hence

$$\begin{aligned} \mathbb{E}_t\Big(w_{t+1}^j\Big) &= \omega\, i_t + b_t^j\ \left(\mathbb{E}_t(s_{t+1}) - s_t + i_t^\star - i_t\right), \\ \mathrm{Var}_t\Big(w_{t+1}^j\Big) &= (b_t^j)^2\, \mathrm{Var}_t(s_{t+1}) \equiv (b_t^j)^2\, \sigma_t^2. \end{aligned}$$

Substituting these, the objective becomes a function of the single choice variable b_t^j:

$$U(b_t^j) = \omega\, i_t + b_t^j\Big[\mathbb{E}_t(s_{t+1}) - s_t(b_t^j) + i_t^\star - i_t\Big] - \frac{\rho}{2}\,(b_t^j)^2\,\sigma_t^2.$$

Differentiating U with respect to b_t^j returns the demand for foreign assets for competitive and strategic investors.

Competitive Investors. A competitive investor takes the exchange rate as given, $\partial s_t/\partial b_t^C = 0$. The first-order condition of $U(b_t^C)$ is

$$\mathbb{E}_t(s_{t+1}) - s_t + i_t^\star - i_t - \rho\, b_t^C\, \sigma_t^2 = 0,$$

which yields the standard mean–variance demand

$$b_t^C = \frac{\mathbb{E}_t(s_{t+1}) - s_t + i_t^\star - i_t}{\rho\, \mathrm{Var}_t(s_{t+1})}.$$

Demand increases with the expected excess return and decreases with risk aversion ρ and the conditional variance σ_t^2.

Strategic Investors. A strategic investor of mass $\lambda_i > 0$ recognizes that its own trade moves the contemporaneous market-clearing exchange rate, $\partial s_t/\partial b_t^{S,i} > 0$, while taking the conditional mean and variance of s_{t+1} as given. Its first-order condition therefore carries an additional term,

$$\mathbb{E}_t(s_{t+1}) - s_t + i_t^\star - i_t - b_t^{S,i}\, \frac{\partial s_t}{\partial b_t^{S,i}} - \rho\, b_t^{S,i}\, \sigma_t^2 = 0,$$

Re-arranging terms:

$$b_t^{S,i} = \frac{\mathbb{E}_t(s_{t+1}) - s_t + i_t^\star - i_t}{\rho\, \mathrm{Var}_t(s_{t+1}) + \frac{\partial s_t}{\partial b_t^{S,i}}}.$$

The strategic schedule differs from the competitive one only through the price-impact term $\partial s_t/\partial b_t^{S,i}$ in the denominator, derived below; setting it to zero recovers b_t^C, so the competitive demand is the price-taking limit. Because the term raises the denominator, a strategic investor holds a smaller position than a competitive one at any given expected excess return.

Step 2: Price Impact

The N strategic investors move simultaneously, each choosing its position to maximize mean–variance utility, taking as given the positions of the other strategic investors, the price-taking demand of the competitive fringe, and noise-trader demand; the exchange rate is set by market clearing, and each strategic investor internalizes the effect of its own position on that rate. The equilibrium exchange rate clears the foreign-bond market,

$$(1-\lambda)\, b_t^C + \sum_{j=1}^{N} \lambda_j\, b_t^{S,j} + (x_t + \bar{x})\bar{W} = B\,(1 + s_t),$$

where the right-hand side linearly approximates Be^{s_t} around $s_t = 0$. This condition implicitly defines the market-clearing rate s_t as a function of investor i's own position $b_t^{S,i}$. Differentiating both sides with respect to $b_t^{S,i}$ — and recalling that the competitive demand responds to $b_t^{S,i}$ only through the price, that the rival strategic quantities are held fixed, and that noise demand is exogenous — gives

$$(1-\lambda)\,\frac{\partial b_t^C}{\partial s_t}\,\frac{\partial s_t}{\partial b_t^{S,i}} + \lambda_i = B\,\frac{\partial s_t}{\partial b_t^{S,i}}.$$

Collecting the terms in $\partial s_t/\partial b_t^{S,i}$,

$$\lambda_i = \left[B - (1-\lambda)\,\frac{\partial b_t^C}{\partial s_t}\right]\frac{\partial s_t}{\partial b_t^{S,i}} \qquad \Longrightarrow \qquad \frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i}{B - (1-\lambda)\,\partial b_t^C/\partial s_t}.$$

The fringe's price sensitivity follows from the competitive schedule: since only $-s_t$ in its numerator depends on the exchange rate,

$$\frac{\partial b_t^C}{\partial s_t} = \frac{\partial}{\partial s_t}\left[\frac{\mathbb{E}_t(s_{t+1}) - s_t + i_t^\star - i_t}{\rho\,\mathrm{Var}_t(s_{t+1})}\right] = -\frac{1}{\rho\,\mathrm{Var}_t(s_{t+1})}.$$

Substituting this and multiplying numerator and denominator by $\rho\,\mathrm{Var}_t(s_{t+1})$,

$$\frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i}{B + \dfrac{1-\lambda}{\rho\,\mathrm{Var}_t(s_{t+1})}} = \frac{\lambda_i\,\rho\,\mathrm{Var}_t(s_{t+1})}{B\,\rho\,\mathrm{Var}_t(s_{t+1}) + (1-\lambda)} \equiv \frac{1}{N}\,\frac{\lambda\,\rho\,\sigma_t^2}{B\,\rho\,\sigma_t^2 + (1-\lambda)} > 0,$$

where the last equality imposes a symmetric oligopoly ($\lambda_i = \lambda/N\ \forall i$). The price impact is positive for all $(B, \lambda, N, \lambda_i, \rho, \sigma_t^2)$, increasing in the investor's mass λ_i and decreasing in the number of strategic investors N, so market power is larger in more concentrated markets. This expression is exactly the slope that appears in each investor's first-order condition.

Step 3: Equilibrium Exchange Rate and Informativeness

Combining the optimal demands with the market-clearing condition delivers the equilibrium exchange rate. Recall the market-clearing condition,

$$(1-\lambda)\,b_t^C + \sum_{i=1}^{N} \lambda_i\,b_t^{S,i} + X_t = B\,(1+s_t), \tag{B.23}$$

and the optimal demands, which are all linear in the common expected excess return $\mathbb{E}_t q_{t+1} \equiv \mathbb{E}_t s_{t+1} - s_t + i_t^\star - i_t$:

$$b_t^C = \frac{\mathbb{E}_t q_{t+1}}{\rho\sigma_t^2}, \qquad b_t^{S,i} = \frac{\mathbb{E}_t q_{t+1}}{\rho\sigma_t^2 + \pi_i}, \qquad \pi_i \equiv \frac{\partial s_t}{\partial b_t^{S,i}} = \frac{\lambda_i\,\rho\sigma_t^2}{B\rho\sigma_t^2 + (1-\lambda)}.$$

Total investor demand is therefore $A\,\mathbb{E}_t q_{t+1}$, where

$$A \equiv \frac{1-\lambda}{\rho\sigma_t^2} + \sum_{i=1}^{N} \frac{\lambda_i}{\rho\sigma_t^2 + \pi_i} \tag{B.24}$$

is the aggregate sensitivity of demand to the expected excess return. Substituting into (B.23) and using $X_t = (\bar{x} + x_t)\bar{W}$,

$$A\left(\mathbb{E}_t s_{t+1} - s_t + i_t^\star - i_t\right) + (\bar{x} + x_t)\bar{W} = B\,(1 + s_t).$$

Collecting the terms in s_t,

$$(A + B)\,s_t = A\,\mathbb{E}_t s_{t+1} + A\,(i_t^\star - i_t) + (\bar{x} + x_t)\bar{W} - B.$$

Divide by $A + B$ and define the *informativeness coefficient*

$$\mu \equiv \frac{A}{A+B}, \tag{B.25}$$

so that $\frac{1}{A+B} = \frac{1-\mu}{B}$. Using $b = B/\bar{W}$ (hence $\bar{W}/B = 1/b$),

$$\begin{aligned} s_t &= \mu\big(\mathbb{E}_t s_{t+1} + i_t^\star - i_t\big) + \frac{1-\mu}{B}\Big[(\bar{x} + x_t)\bar{W} - B\Big] \\ &= \mu\big(\mathbb{E}_t s_{t+1} + i_t^\star - i_t\big) + (1-\mu)\Big[\frac{\bar{x} + x_t}{b} - 1\Big] = \underbrace{(1-\mu)\Big(\frac{\bar{x}}{b} - 1\Big)}_{\text{constant}} + \underbrace{\mu\big(\mathbb{E}_t s_{t+1} + i_t^\star - i_t\big)}_{\text{fundamental}} + \underbrace{(1-\mu)\frac{1}{b}x_t}_{\text{noise}}, \end{aligned}$$

which is the equilibrium law of motion of the exchange rate.

Closed form for μ. The aggregate demand sensitivity A in (B.24) admits a closed form for an arbitrary distribution of strategic masses $\{\lambda_i\}_{i=1}^N$, without any symmetry assumption. Using the price impact $\pi_i = \lambda_i\rho\sigma_t^2/\big(B\rho\sigma_t^2 + (1-\lambda)\big)$ and writing $V \equiv B\rho\sigma_t^2$ and $D \equiv V + (1-\lambda)$, each denominator satisfies $\rho\sigma_t^2 + \pi_i = \rho\sigma_t^2\,\frac{D+\lambda_i}{D}$, so

$$A = \frac{1-\lambda}{\rho\sigma_t^2} + \sum_{i=1}^{N} \frac{\lambda_i}{\rho\sigma_t^2 + \pi_i} = \frac{1}{\rho\sigma_t^2}\left[(1-\lambda) + D\sum_{i=1}^{N}\frac{\lambda_i}{D+\lambda_i}\right] \equiv \frac{S}{\rho\sigma_t^2}.$$

Substituting into (B.25) and using $B\rho\sigma_t^2 = V$ gives the closed form

$$\mu = \frac{A}{A+B} = \frac{S}{S+V}, \qquad S \equiv (1-\lambda) + D\sum_{i=1}^{N} \frac{\lambda_i}{D+\lambda_i}, \quad V \equiv B\rho\,\mathrm{Var}_t(s_{t+1}). \tag{B.26}$$

The symmetry assumption allows us to express μ compactly in terms of (λ, N). In the symmetric case $\lambda_i = \lambda/N$, the sum collapses to $\sum_i \frac{\lambda_i}{D+\lambda_i} = \frac{\lambda N}{ND+\lambda}$, so $S = \frac{ND+\lambda-\lambda^2}{ND+\lambda}$; writing $ND + \lambda = N\bigl(1 + V - \lambda\frac{N-1}{N}\bigr) \equiv NG$ with $G \equiv 1 + V - \lambda\frac{N-1}{N} > 0$,

$$\mu = \frac{1 + V - \lambda\frac{N-1}{N} - \frac{\lambda^2}{N}}{(1+V)\bigl(1 + V - \lambda\frac{N-1}{N}\bigr) - \frac{\lambda^2}{N}}, \qquad V \equiv B\rho\,\mathrm{Var}_t(s_{t+1}). \tag{B.27}$$

In both cases $\mu \in (0,1)$ because $A > 0$.

Forward-looking representation. Let the constant term be defined as:

$$c \equiv (1-\mu)\left(\frac{\bar{x}}{b} - 1\right)$$

Rewriting s_t by separating the current expectations and exogenous shocks:

$$s_t = c + \frac{1-\mu}{b}x_t + \mu(i_t^\star - i_t) + \mu\mathbb{E}_t[s_{t+1}] \tag{B.28}$$

Forwarding Equation (B.28) by one period to period $t+1$:

$$s_{t+1} = c + \frac{1-\mu}{b}x_{t+1} + \mu(i_{t+1}^\star - i_{t+1}) + \mu\mathbb{E}_{t+1}[s_{t+2}]$$

Taking the conditional expectation at time t ($\mathbb{E}_t[\cdot]$) and applying the Law of Iterated Expectations ($\mathbb{E}_t[\mathbb{E}_{t+1}[\cdot]] = \mathbb{E}_t[\cdot]$):

$$\mathbb{E}_t[s_{t+1}] = c + \frac{1-\mu}{b}\mathbb{E}_t[x_{t+1}] + \mu\mathbb{E}_t[i_{t+1}^\star - i_{t+1}] + \mu\mathbb{E}_t[s_{t+2}]$$

Substituting this expression back into Equation (B.28):

$$\begin{aligned} s_t &= c + \frac{1-\mu}{b}x_t + \mu(i_t^\star - i_t) + \mu\left(c + \frac{1-\mu}{b}\mathbb{E}_t[x_{t+1}] + \mu\mathbb{E}_t[i_{t+1}^\star - i_{t+1}] + \mu\mathbb{E}_t[s_{t+2}]\right) \\ &= c(1+\mu) + \frac{1-\mu}{b}\left(x_t + \mu\mathbb{E}_t[x_{t+1}]\right) + \mu(i_t^\star - i_t) + \mu^2\mathbb{E}_t[i_{t+1}^\star - i_{t+1}] + \mu^2\mathbb{E}_t[s_{t+2}] \end{aligned}$$

Iterating forward T periods yields:

$$s_t = c\sum_{k=0}^{T-1}\mu^k + \frac{1-\mu}{b}\sum_{k=0}^{T-1}\mu^k\mathbb{E}_t[x_{t+k}] + \sum_{k=0}^{T-1}\mu^{k+1}\mathbb{E}_t\left[i^\star_{t+k} - i_{t+k}\right] + \mu^T\mathbb{E}_t[s_{t+T}]$$

Assuming $0 < \mu < 1$, we impose the standard transversality condition:

$$\lim_{T\to\infty}\mu^T\mathbb{E}_t[s_{t+T}] = 0$$

Evaluating the geometric series for the constant term as $T \to \infty$:

$$c\sum_{k=0}^{\infty}\mu^k = \frac{c}{1-\mu} = \frac{(1-\mu)\left(\frac{\bar{x}}{b}-1\right)}{1-\mu} = \frac{\bar{x}}{b} - 1$$

Taking the limit $T \to \infty$ produces the full unrolled forward-looking representation:

$$s_t = \left(\frac{\bar{x}}{b} - 1\right) + \sum_{k=0}^{\infty}\mu^{k+1}\mathbb{E}_t\left[i^\star_{t+k} - i_{t+k}\right] + \frac{1-\mu}{b}\sum_{k=0}^{\infty}\mu^k\mathbb{E}_t\left[x_{t+k}\right] \tag{B.29}$$

B.2 Proofs of the Theoretical Results

Lemma 1 *Holding the conditional variance σ_t^2 fixed, μ is strictly decreasing in the size of the strategic segment λ and strictly increasing in the number of strategic investors N.*

Proof. We consider the symmetric case, in which μ is given by (B.27). Write $G \equiv 1+V-\lambda\frac{N-1}{N} > 0$ and $h \equiv \lambda^2/(NG)$, so that

$$\mu = \frac{1}{1+\dfrac{V}{1-h}} = \frac{1-h}{1-h+V}.$$

Holding σ_t^2 (hence $V = B\rho\sigma_t^2$) fixed,

$$\frac{\partial\mu}{\partial h} = \frac{-(1-h+V)+(1-h)}{(1-h+V)^2} = -\frac{V}{(1-h+V)^2} < 0,$$

so μ is strictly decreasing in h; by the chain rule, $\partial\mu/\partial\lambda$ and $\partial\mu/\partial N$ carry the opposite sign to $\partial h/\partial\lambda$ and $\partial h/\partial N$. Since G depends on λ through $\partial G/\partial\lambda = -\frac{N-1}{N}$, differentiating $h = \frac{1}{N}\frac{\lambda^2}{G}$ gives

$$\frac{\partial h}{\partial\lambda} = \frac{2\lambda}{NG} - \frac{\lambda^2}{NG^2}\frac{\partial G}{\partial\lambda} = \frac{2\lambda}{NG} + \frac{\lambda^2(N-1)}{N^2G^2} > 0,$$

so $\partial\mu/\partial\lambda < 0$. Treating N as continuous, $\partial G/\partial N = -\lambda/N^2$, hence $\partial(NG)/\partial N = G - \lambda/N = 1+V-\lambda$ and

$$\frac{\partial h}{\partial N} = -\lambda^2\frac{\partial(NG)/\partial N}{(NG)^2} = -\frac{\lambda^2\,(1+V-\lambda)}{(NG)^2} < 0,$$

where $1+V-\lambda > 0$ because $\lambda < 1$; hence $\partial\mu/\partial N > 0$. Therefore μ is strictly decreasing in λ and strictly increasing in N.

∎

Proposition 1 *An increase in the size of the strategic segment λ amplifies the response of the exchange rate to noise shocks and dampens its response to fundamental shocks.*

Proof. Consider the law of motion of the exchange rate in Equation (12):

$$s_t = -\mu\sum_{k=0}^{\infty}\mu^k\mathbb{E}_t\left(\Delta i_{t+k}\right) + \frac{1-\mu}{b}\sum_{k=0}^{\infty}\mu^k\mathbb{E}_t\left(x_{t+k}\right),$$

where $\Delta i_{t+k} = i_{t+k} - i^\star_{t+k}$. This expresses the spot rate as a discounted sum of *expected* future fundamentals and noise.

The interest differential and the noise component follow the $AR(1)$ processes in Equations (2–3) and (7), so under rational expectations the model-consistent forecast k periods ahead is

$$\mathbb{E}_t\left(\Delta i_{t+k}\right) = \rho_u^k \Delta i_t, \qquad \mathbb{E}_t\left(x_{t+k}\right) = \rho_x^k x_t$$

Substituting these expectations back into the structural equation for the exchange rate:

$$s_t = -\mu \sum_{k=0}^{\infty} \mu^k \rho_u^k \Delta i_t + \frac{1-\mu}{b} \sum_{k=0}^{\infty} \mu^k \rho_x^k x_t,$$

Because $|\rho_u \mu| < 1$ and $|\rho_x \mu| < 1$, evaluating the geometric series yields the reduced-form policy function:

$$s_t = -\left(\frac{\mu}{1-\mu\rho_u}\right)\Delta i_t + \left(\frac{1-\mu}{b(1-\mu\rho_x)}\right) x_t$$

Since it holds in every period, and Δi_t and x_t inherit the $AR(1)$ innovations $\varepsilon_t^u, \varepsilon_t^x$ (standardized, so that a one-standard-deviation shock corresponds to an innovation of σ_u or σ_x), the response of the exchange rate to a one-standard-deviation shock $h \geq 0$ periods after impact is

$$\mathrm{IRF}_s^{\Delta i}(h) = -\frac{\mu}{1-\mu\rho_u}\,\rho_u^h\,\sigma_u, \qquad \mathrm{IRF}_s^{x}(h) = \frac{1-\mu}{b\,(1-\mu\rho_x)}\,\rho_x^h\,\sigma_x, \tag{B.30}$$

The horizon factors $\rho_u^h \sigma_u$ and $\rho_x^h \sigma_x / b$ are positive and independent of μ, so taking the derivative of these responses with respect to μ gives

$$\frac{\partial\left|\mathrm{IRF}_s^{\Delta i}(h)\right|}{\partial\mu} = \frac{\rho_u^h\,\sigma_u}{(1-\mu\rho_u)^2} > 0, \qquad \frac{\partial\,\mathrm{IRF}_s^{x}(h)}{\partial\mu} = -\frac{(1-\rho_x)\,\rho_x^h\,\sigma_x}{b\,(1-\mu\rho_x)^2} < 0. \tag{B.31}$$

Since μ is a decreasing (increasing) function of λ (N) by Lemma 1, an increase in λ (a decrease in N) dampens the response of the exchange rate to fundamental shocks while noise shocks are amplified, at every horizon $h \geq 0$.

■

Proposition 2. *The Fama coefficient in the regression of the realized excess currency return on the interest differential is* $\beta^{\text{Fama}} = -\dfrac{1-\mu}{1-\mu\rho_u} \in (-1, 0)$, *and it becomes more negative as* λ *increases.*

Proof. Define the one-period realized excess return $q_{t+1} = s_{t+1} - s_t - (i_t - i_t^\star)$ and consider the Fama regression

$$q_{t+1} = \alpha + \beta_1\,(i_t - i_t^\star) + \epsilon_{t+1}, \qquad \beta_1 = \frac{\mathrm{Cov}\big(q_{t+1}, \Delta i_t\big)}{\mathrm{Var}(\Delta i_t)}, \quad \Delta i_t \equiv i_t - i_t^\star.$$

From the law of motion of the exchange rate,

$$s_t = \mu\big[\mathbb{E}_t(s_{t+1}) + i_t^\star - i_t\big] + (1-\mu)\frac{\bar{x}}{b} + (1-\mu)\frac{1}{b}\,x_t,$$

only the first term loads on fundamentals; the noise component is orthogonal to the interest differential and does not covary with Δi_t. Iterating forward under rational expectations and using the $AR(1)$ forecast $\mathbb{E}_t(\Delta i_{t+m}) = \rho_u^m \Delta i_t$, the *expected* j-period change in the exchange rate is

$$\mathbb{E}_t\,\Delta s_{t+j} = -\mu \sum_{k=0}^{\infty} \mu^k\,\mathbb{E}_t\big(\Delta i_{t+j+k} - \Delta i_{t+k}\big). \tag{B.32}$$

Because forecast errors are orthogonal to the time-t information set, and hence to Δi_t, we have $\mathrm{Cov}(\Delta s_{t+j}, \Delta i_t) = \mathrm{Cov}(\mathbb{E}_t \Delta s_{t+j}, \Delta i_t)$: the realized change may be replaced by its conditional expectation (B.32) throughout. Taking $j = 1$,

$$\begin{aligned}
\beta_1 &= \frac{\mathrm{Cov}\big(\Delta s_{t+1} - \Delta i_t,\ \Delta i_t\big)}{\mathrm{Var}(\Delta i_t)} = \frac{1}{\mathrm{Var}(\Delta i_t)}\left[\mathrm{Cov}\Big(-\mu\sum_{k=0}^{\infty}\mu^k\big(\Delta i_{t+k+1} - \Delta i_{t+k}\big),\ \Delta i_t\Big) - \mathrm{Var}(\Delta i_t)\right] \\
&= \frac{1}{\mathrm{Var}(\Delta i_t)}\left[-\mu\sum_{k=0}^{\infty}\mu^k\,\mathrm{Cov}\big(\Delta i_{t+k+1} - \Delta i_{t+k},\ \Delta i_t\big) - \mathrm{Var}(\Delta i_t)\right] \\
&= -\mu\sum_{k=0}^{\infty}\mu^k\rho_u^k(\rho_u - 1) - 1 = -\mu\,\frac{\rho_u - 1}{1-\mu\rho_u} - 1 = -\frac{1-\mu}{1-\mu\rho_u} \ \in (-1, 0),
\end{aligned}$$

which is negative for every μ and increasing in μ (hence decreasing in λ).

■

Predictability j periods ahead. Predictability reversal does not arise in our model, unlike in Bacchetta and Van Wincoop (2010) and Engel (2016). Define the excess return realized over $[t+j-1, t+j]$,

$$q_{t+j} = s_{t+j} - s_{t+j-1} - (i_{t+j-1} - i_{t+j-1}^\star),$$

and the regression $q_{t+j} = \alpha + \beta_j\,(i_t - i_t^\star) + \epsilon_{t+j}$. Proceeding exactly as above—again replacing realized changes by their time-t expectations—

$$\begin{aligned}
\beta_j &= \frac{\mathrm{Cov}(q_{t+j}, \Delta i_t)}{\mathrm{Var}(\Delta i_t)} = \frac{1}{\mathrm{Var}(\Delta i_t)}\Big[\mathrm{Cov}(\Delta s_{t+j}, \Delta i_t) - \mathrm{Cov}(\Delta i_{t+j-1}, \Delta i_t)\Big] \\
&= -\mu\sum_{k=0}^{\infty}\mu^k\big(\rho_u^{k+j} - \rho_u^{k+j-1}\big) - \rho_u^{j-1} = -\mu\,\rho_u^{j-1}\frac{\rho_u - 1}{1-\mu\rho_u} - \rho_u^{j-1} = -\rho_u^{j-1}\frac{1-\mu}{1-\mu\rho_u} \ \leq 0.
\end{aligned}$$

Since $0 \le \rho_u < 1$, successive coefficients satisfy $\beta_{j+1} = \rho_u \beta_j$, so $|\beta_j|$ decays geometrically and $\beta_j \uparrow 0$ monotonically as $j \to \infty$ without ever changing sign: no predictability reversal occurs.[20]

Proposition 3. *The unconditional variance of the exchange rate is*

$$\mathrm{Var}(s_t) = \frac{\mu^2}{(1-\mu\rho_u)^2}\,\frac{\sigma_u^2}{1-\rho_u^2} + \frac{(1-\mu)^2}{(1-\mu\rho_x)^2\, b^2}\,\frac{\sigma_x^2}{1-\rho_x^2}.$$

Proof. The equilibrium law of motion admits the reduced form (Proposition 1),

$$s_t = c_u\,\Delta i_t + c_x\, x_t, \qquad c_u = -\frac{\mu}{1-\mu\rho_u}, \quad c_x = \frac{1-\mu}{(1-\mu\rho_x)\, b}, \tag{B.33}$$

so the spot rate is a function of the *current* states Δi_t and x_t only. The two states are mutually orthogonal $AR(1)$ processes with

$$\mathrm{Var}(\Delta i_t) = \frac{\sigma_u^2}{1-\rho_u^2}, \qquad \mathrm{Var}(x_t) = \frac{\sigma_x^2}{1-\rho_x^2}.$$

Since $\mathrm{Cov}(\Delta i_t, x_t) = 0$, the variance of (B.33) contains no cross term, so

$$\mathrm{Var}(s_t) = c_u^2\,\mathrm{Var}(\Delta i_t) + c_x^2\,\mathrm{Var}(x_t) = \frac{\mu^2}{(1-\mu\rho_u)^2}\,\frac{\sigma_u^2}{1-\rho_u^2} + \frac{(1-\mu)^2}{(1-\mu\rho_x)^2\, b^2}\,\frac{\sigma_x^2}{1-\rho_x^2}. \tag{B.34}$$

■

Corollary 1. *An increase in the size of strategic investors makes the exchange rate more volatile relative to fundamentals if and only if the noise-to-fundamental variance ratio exceeds the threshold $\bar{\Omega}$.*

$$\frac{\mathrm{Var}(x_t)}{\mathrm{Var}(i_t - i_t^\star)} > \overline{\Omega}, \qquad \overline{\Omega} \equiv b^2\,\frac{\mu\,(1-\mu\rho_x)^3}{(1-\mu)\,(1-\rho_x)\,(1-\mu\rho_u)^3}.$$

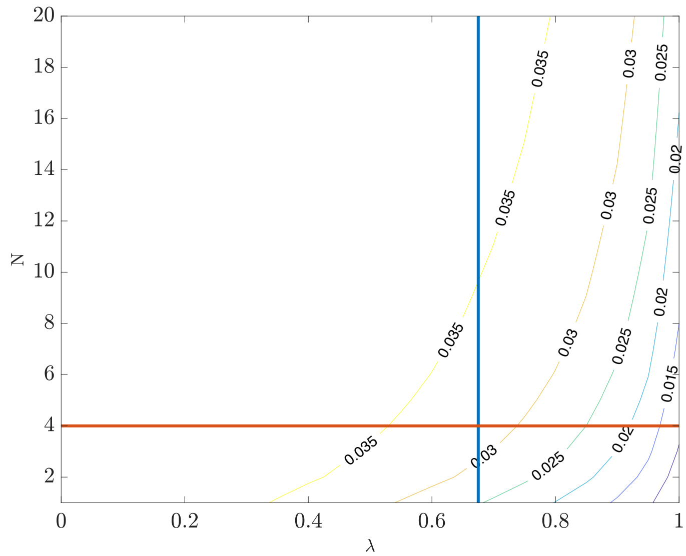
Proof. Differentiating (B.34) with respect to μ,

$$\frac{\partial\,\mathrm{Var}(s_t)}{\partial\mu} = \frac{2\mu}{(1-\mu\rho_u)^3}\,\frac{\sigma_u^2}{1-\rho_u^2} - \frac{2(1-\mu)(1-\rho_x)}{(1-\mu\rho_x)^3}\,\frac{\sigma_x^2}{b^2\,(1-\rho_x^2)}. \tag{B.35}$$

By Lemma 1, μ is decreasing in λ; hence exchange-rate volatility increases with λ if and only if $\partial\,\mathrm{Var}(s_t)/\partial\mu < 0$. Setting the second term of (B.35) above the first and rearranging (using

[20] This is not surprising given the absence of any friction, such as infrequent portfolio adjustment (Bacchetta and Van Wincoop, 2010, 2019).

Figure B.6: $\underline{\sigma_x}$ for different combinations of N and λ.



Notes: The figure shows the minimum value of the volatility of the noise process, σ_x, that guarantees that the volatility of the exchange rate is monotonically increasing in the presence of strategic behavior (higher λ and/or lower N). We compute the minimum value of σ_x for different levels of λ and N. The horizontal and vertical lines pin down the combination of λ and N used in the parametrization of the basic framework. Remaining parameters are constant, see Table 2.

$\text{Var}(\Delta i_t) = \sigma_u^2/(1-\rho_u^2)$, $\text{Var}(x_t) = \sigma_x^2/(1-\rho_x^2))$ gives the threshold condition

$$\frac{\text{Var}(x_t)}{\text{Var}(i_t - i_t^\star)} > \overline{\Omega}, \qquad \overline{\Omega} \equiv b^2 \frac{\mu\,(1-\mu\rho_x)^3}{(1-\mu)\,(1-\rho_x)\,(1-\mu\rho_u)^3}. \tag{B.36}$$

■

Condition (B.36) can be expressed as a lower bound on the noise volatility. Writing it with equality and solving for σ_x delivers

$$\underline{\sigma_x} = b\,\sigma_u \sqrt{\frac{\mu\,(1-\mu\rho_x)^3\,(1+\rho_x)}{(1-\mu)\,(1-\mu\rho_u)^3\,(1-\rho_u^2)}}\;, \tag{B.37}$$

the minimum noise volatility above which exchange-rate volatility rises with strategic behavior. Since $\mu = \mu(\lambda, N)$ (Lemma 1), $\underline{\sigma_x}$ is a function of the market structure; Figure B.6 plots it across (λ, N).

The non-monotonic case does not arise under standard parametrizations, including ours. In our calibration, σ_x would have to be at least 75% below its benchmark value to overturn the monotonic relationship between strategic behavior (λ and/or N) and the unconditional variance of the exchange rate; for other combinations of (λ, N) the required reduction is at least 50% (Figure C.8). For instance, at a high level of strategic behavior ($\lambda \approx 1$) the benchmark noise volatility is $\sigma_x \approx 0.05$, and monotonicity breaks only if σ_x falls below 0.025. This robustness is reinforced by our conservative calibration: $\underline{\sigma}_x$ decreases in the noise persistence ρ_x and in home bias (b^{-1}) and increases in the fundamental persistence ρ_u, so the parameter movements consistent with the data—more persistent noise, less persistent fundamentals, or stronger home bias—all push the threshold lower and relax the condition.

Proposition 4. *The share of the variance of exchange-rate changes explained by fundamentals is*

$$R^2 = \frac{\text{Var}(i_t - i_t^\star)}{\text{Var}(\Delta s_{t+1})} \left[1 + \beta^{Fama}\right]^2. \tag{B.38}$$

Proof. Define the exchange rate change as $\Delta s_{t+1} = s_{t+1} - s_t$ and the interest rate differential as Δi_t. The exchange rate disconnect can be measured by the R^2 of the following regression:

$$\Delta s_{t+1} = \alpha + \beta \Delta i_t + \varepsilon_t$$

The R^2 is given by:

$$R^2 = \text{Corr}(\Delta s_{t+1}, \Delta i_t)^2 = \frac{\text{Cov}(\Delta s_{t+1}, \Delta i_t)^2}{\text{Var}(\Delta s_{t+1})\,\text{Var}(\Delta i_t)} = \left[\frac{\text{Cov}(\Delta s_{t+1}, \Delta i_t)}{\text{Var}(\Delta i_t)}\right]\left[\frac{\text{Cov}(\Delta s_{t+1}, \Delta i_t)}{\text{Var}(\Delta s_{t+1})}\right]. \tag{B.39}$$

Adding and subtracting $\text{Var}(\Delta i_t)$ in the numerator, we have:

$$\begin{aligned} R^2 &= \left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t)}{\text{Var}(\Delta i_t)} + 1\right]\left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t) + \text{Var}(\Delta i_t)}{\text{Var}(\Delta s_{t+1})}\right], \\ &= \left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t)}{\text{Var}(\Delta i_t)} + 1\right]\frac{\text{Var}(\Delta i_t)}{\text{Var}(\Delta s_{t+1})}\left[\frac{\text{Cov}(\Delta s_{t+1} - \Delta i_t, \Delta i_t)}{\text{Var}(\Delta i_t)} + 1\right]. \end{aligned}$$

Since the covariance term is the Fama coefficient β^{Fama}, it follows that:

$$R^2 = \frac{\text{Var}(\Delta i_t)}{\text{Var}(\Delta s_{t+1})}\left[1 + \beta^{Fama}\right]^2. \tag{B.40}$$

■

Corollary 2. *The disconnect between the exchange rate and fundamentals increases in the size of strategic investors.*

Proof. We show that $\frac{\partial R^2}{\partial \mu} > 0$ for any stationary parametrization with $\sigma_x > 0$. Since μ is strictly decreasing in λ (Lemma 1), this is equivalent to R^2 being strictly decreasing in λ, which is the content of Corollary 2.

From the closed-form solution $s_t = c_u\,\Delta i_t + c_x\,x_t$ (Proposition 1), with $c_u = -\mu/(1-\mu\rho_u)$ and $c_x = (1-\mu)/[(1-\mu\rho_x)b]$, the Fama coefficient is $\beta^{Fama} = -(1-\mu)/(1-\mu\rho_u)$, so $1+\beta^{Fama} = \mu(1-\rho_u)/(1-\mu\rho_u)$. Using the orthogonality of Δi_t and x_t and $\mathrm{Var}(z_{t+1}-z_t) = 2\sigma_z^2/(1+\rho_z)$ for an $AR(1)$,

$$\mathrm{Var}(\Delta s_{t+1}) = 2\,c_u^2\frac{\sigma_u^2}{1+\rho_u} + 2\,c_x^2\frac{\sigma_x^2}{1+\rho_x} \equiv 2\,(F+\mathcal{N}), \qquad F \equiv c_u^2\frac{\sigma_u^2}{1+\rho_u},\ \ \mathcal{N} \equiv c_x^2\frac{\sigma_x^2}{1+\rho_x}.$$

Since $[1+\beta^{Fama}]^2\,\mathrm{Var}(\Delta i_t) = c_u^2(1-\rho_u)^2\sigma_u^2/(1-\rho_u^2) = (1-\rho_u)\,F$, the R^2 becomes

$$R^2 = \frac{1-\rho_u}{2}\,\frac{F}{F+\mathcal{N}} = \frac{1-\rho_u}{2}\,\frac{1}{1+\Psi}, \qquad \Psi \equiv \frac{\mathcal{N}}{F} = \frac{(1-\mu)^2}{\mu^2}\,\frac{(1-\mu\rho_u)^2}{(1-\mu\rho_x)^2}\,\frac{1}{b^2}\,\frac{\sigma_x^2/(1+\rho_x)}{\sigma_u^2/(1+\rho_u)}.$$

Thus R^2 is decreasing in the noise contribution Ψ. Differentiating, the sign of $\partial R^2/\partial\mu$ equals that of

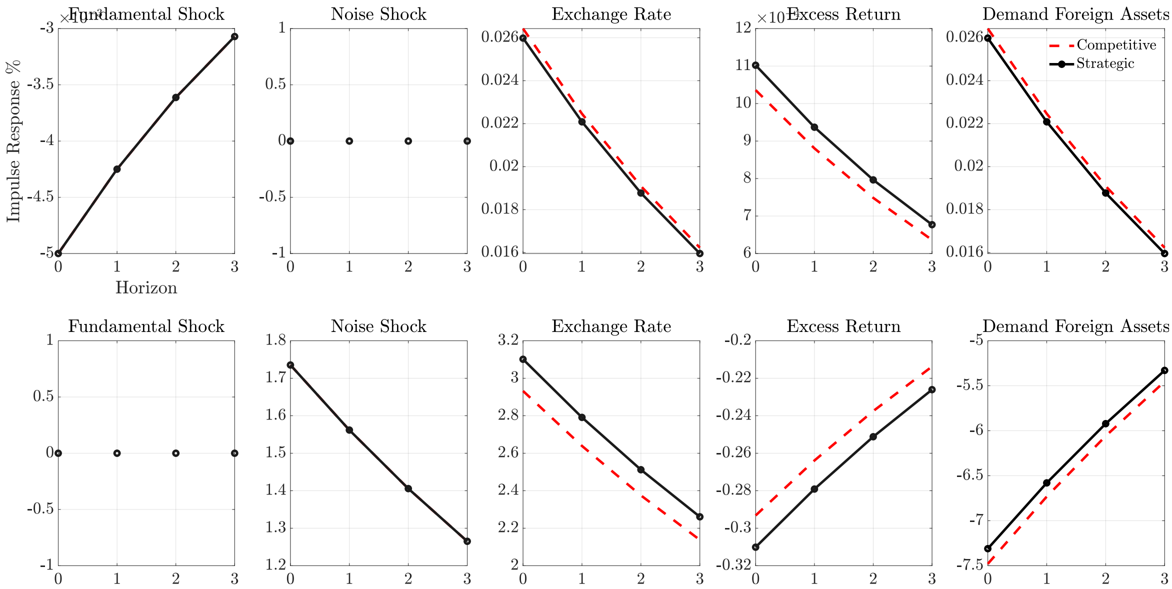
$$Q(\mu) \equiv 1 - 2\mu\rho_x + \mu^2\big(\rho_x + \rho_u\rho_x - \rho_u\big),$$

which is decreasing in ρ_u and therefore bounded below by its value at $\rho_u = 1$, $Q(\mu) > (1-\mu)(1+\mu-2\mu\rho_x) > 0$. Hence $\partial R^2/\partial\mu > 0$: R^2 is increasing in μ. By Lemma 1, μ is decreasing in the size of the strategic segment λ (and increasing in N), so R^2 is decreasing in λ. The disconnect between the exchange rate and fundamentals therefore increases in the size of strategic investors.

■

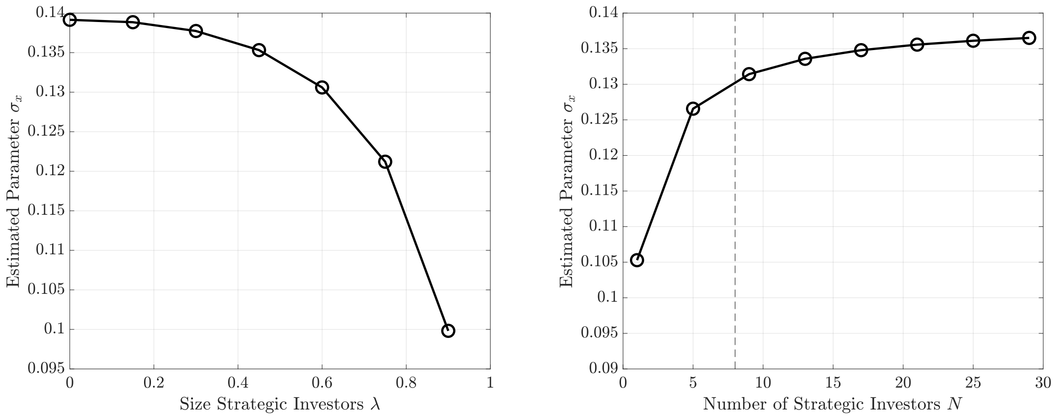
C Quantitative Assessment

Figure C.7: Impulse Response to Exogenous Shocks



Notes: The top panel (bottom) shows the response to a two-standard deviation shock in fundamental and noise. The first and second columns show the dynamics of the exogenous shocks in fundamentals and noise, respectively. The third column shows the dynamics of the exchange rate. Column four shows the response of the realized excess return, defined as $q_{t+1} = s_{t+1} - s_t + i_t^{\star} - i_t$. The last column shows the response of the total demand of foreign assets, defined as $(1-\lambda)b_t^C + \sum_i^N \lambda_i b_t^{S,i}$, where b_t^C and $b_t^{S,i}$ are defined according to Equation 10. The solid black line shows the response in the benchmark parametrization with strategic investors, $\lambda = 0.6$. The red dashed line shows the response in a competitive economy without strategic investors, $\lambda = 0$. Remaining parameters are common across scenarios, see Table 2.

Figure C.8: Relationship between Strategic Behavior and Noise Volatility



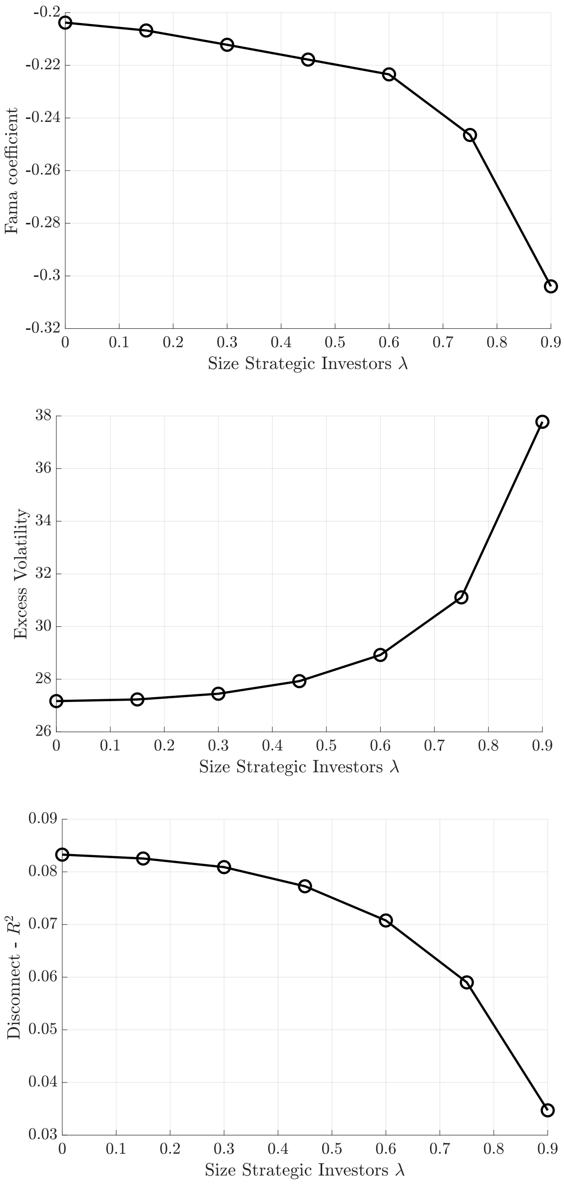
Notes: The figure shows the volatility of the noise component, σ_x, required to match the target volatility of the exchange rate change in the basic framework, for different levels of strategic behavior. The left panel considers different levels of strategic behavior in terms of λ for a number of strategic investors equal to $N = 4$. The left panel considers different levels of strategic behavior in terms of N for a total size of strategic investors equal to $\lambda = 0.675$. All other parameters are constant and summarized in Table 2.

Table C.7: Potential Impact of Strategic Investors in Benchmark Model

Panel A: Moments from Simulated Exchange Rate Data	(1) Response to Shock Noise (Fundamental)	(2) Fama Coefficient	(3) Exchange Rate Volatility (St.D.)	(4) Exchange Rate Disconnect
Competitive (Atomistic investors)	14.66 (-0.088)	-0.204	0.259	0.083
Strategic (Avg. investor 7%)	15.506 (-0.087)	-0.217	0.274	0.073
Periods per Simulation	6000	6000	6000	6000
Burn-in Periods	3000	3000	3000	3000
Number of Simulations	1000	1000	1000	1000
Panel B: Moments from Actual Exchange Rate Data		Fama Coefficient	Exchange Rate Volatility (St.D.)	Exchange Rate Disconnect
Monthly Avg. Across Currencies		-0.603	0.344	0.030
(St.D.)		(1.317)	(0.353)	(0.065)
Weekly Avg. Across Currencies		-0.490	0.341	0.036
(St.D.)		(0.780)	(0.339)	(0.064)
Years per Currency		1993-2019	1993-2019	1993-2019
Number of Currencies		18	18	18

Notes: Panel A reports the impact of strategic investors on exchange rate dynamics using simulated data, with results for the competitive and benchmark models in the first and second rows, respectively. Column (1) shows the exchange rate response to a one-standard-deviation noise and fundamental shock, expressed as a percentage deviation from the steady state (e.g., 1 indicates a 1% deviation). Column (2) displays the Fama regression beta, indicating deviations from uncovered interest parity. Column (3) reports exchange rate volatility (standard deviation), and Column (4) provides the estimated R^2 from the disconnect regression in the simulated data. Panel B shows the average and standard deviation of equivalent moments from actual exchange rate data at monthly and weekly frequencies. The results in Panel A are based on 1,000 simulations, each running for 6,000 periods with a 3,000-period burn-in. All other parameters are constant across scenarios; see Table 2 for details. Appendix A provides further data information, and Appendix C outlines the estimation procedure.

Figure C.9: Exchange Rate Disconnect and Excess Volatility



Notes: The figure shows the excess predictability coefficient from regression in Equation 16 (top), the excess volatility ratio (middle), and the estimated R^2 of the disconnect regression in Equation 21 (bottom) using simulated data. We run 5,000 simulations and, for each iteration, the model runs for 8,000 periods with 3,000 burn-in. Data are simulated for different levels of strategic behavior λ. Remaining parameters are common across scenarios, see Table 2.